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A meta-analysis of the observational studies: the correlation between masked hypertension and endothelial dysfunction measured using flow-mediated dilation

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A meta-analysis of the observational studies: the correlation between masked hypertension and endothelial dysfunction measured using flow-mediated dilation Abstract

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Keywords: blood pressure, masked hypertension, flow-mediated dilation, meta, protocol

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Introduction

A surrogate marker to evaluate the endothelial response of the artery when stimulated by reactive hyperemia, known as brachial flow-mediated dilation(FMD), have prognostic value to predict hypertensive organ damage and cardiovascular disease(CVD) events. However, the degree of correlation between brachial artery FMD and masked hypertension outcomes remains less clear. This study's purpose is to pool together data regarding FMD with respect to masked hypertension.

Methods and analysis

Electronic databases (MEDLINE, EMBASE, and the Cochrane Library) will be searched for the keywords hypertension, masked hypertension, flow-mediated dilation. Authors and reference inventory of included studies also will be contacted to identify additional published and unpublished studies. Eligibility criteria are as shown: population—adults (over 18 years old or older with no upper limit) at baseline, from hypertensive groups (systolic blood pressure $\geq$140 mmHg and/or diastolic BP $\geq$90 mmHg or take antihypertensive drugs for hypertension) and from primary care, community cohort, electronic database registry, or randomized controlled trial; exposure—any metric of FMD, ambulatory BP monitoring, home BP monitoring, office BP; comparison—persons without masked hypertension who do not have elevated BP; and outcome—masked hypertension; Two writers will be engaged in screening and collecting data independently, and disagreements resolved by discussing. Data extraction will include but not limited to basic data designated as hazard ratios, odds ratios, correlations, regression coefficients, and original cell data if available. We will use Comprehensive Meta-Analysis Version 2.0 to conduct the meta-analyses.

Ethics and dissemination

This study does not require ethics approval. It will be disseminated internationally and presented in the peer-reviewed literature.

PROSPERO registration number: CRD42020208362

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Background

Hypertension is a high morbidity disease[1], leading to seriously threaten the cardiovascular and renal system, and comes at heavy economic burdens. Masked hypertension(MH), is defined as having a normal blood pressure in office despite having an elevated measurement outside of it[2]. The diagnosis of masked hypertension (MH) remains challenging. Both missed diagnosis and excess diagnosis have undesirable consequences. Undiagnosed MH could be frequently developed target organ damage prior to transitioning to sustained hypertension[3]. Moreover, there is a found that taking antihypertensive medication has been found to have an increased prevalence rate of masked hypertension as compared to those not taking antihypertensive medication[4]. A definitive and accurate diagnostic conclusion is therefore required for all patients with suspected MH. Although studies have shown that hypertension is associated with increased risk of CVD, the actual concern about this health risk is not the mild elevation in BP measured in the clinic, but the underestimation of masked hypertension.

The diagnostic strategy starts with clinical probability assessment and blood pressure (BP) in the clinic. In patients with a high/likely clinical probability, ambulatory BP monitoring (ABPM) and home BP monitoring (HBPM) is required. ABPM is currently the predominantly used test for MH. The test has been widely verified in diagnostic accuracy and outcome studies[5-7]. However, ABPM has some limitations, including substantial disagreement when versus HBPM[8], hard to operate, and a continuous concern regarding a possible over diagnosis and overtreatment of hypertension.

In recent years, the technology around quantifying endothelial function has rapidly evolved allowing the introduction of flow-mediated dilation, a non-invasive method of assessing endothelial dysfunction[9]. Flow-mediated dilation(FMD) has been reported to predict hypertensive organ damage and CVD events[10 11]. FMD has already largely replaced using the invasive method during cardiac catheterization. Many studies aimed at assessing the correlation between FMD and CVD risk. More recently, a few systematic reviews and meta-analyses have been published, and conclude that FMD was positively associated with risk of future cardiovascular events[12 13]. However, the clinical community remains unclear about the relationships between masked hypertension diagnosed through ABPM and flow‐mediated vasodilation response. Indeed, the concrete relations of the FMD and masked hypertension is still unclear. For example, in some studies, there is a bias with the FMD results participating in the ultimate diagnosis conclusion. It is important to notice that the previously published meta-analyses of brachial FMD have been reported for the outcomes of CVD. This is an important limitation that needs to be encountered when interpreting summary results. Besides, there is extensive heterogeneity in the studies according to reference standard and criteria used for interpretation. Thus, the clinical outcome of patients measured through FMD remains unclear.

Given there is no data available on the assessment of endothelial function measured by FMD in subject with masked-hypertension, there is an urgent necessity to perform

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a systematic review and meta-analysis on the performances of FMD for diagnosis of MH, using objective and widely accepted tools for quality assessment.

Methods

Aims

The study aims to synthesize the evidence base regarding FMD and masked hypertension. The reporting of this study protocol conforms to the PRISMA-P guidelines (supplementary appendix 1) and was registered on PROSPERO database[14]. The full study will accord with the PRISMA guidelines. Updates to this study will be registered on PROSPERO.

Search strategy

We will search for relevant articles in English and Chinese by these databases: MEDLINE, EMBASE, the Cochrane Library and Chinese National Knowledge Infrastructure(CNKI). The search period will be from the date of database inception to November 2020. We will perform a hand search of the reference lists of articles selected to supplement the electronic search. We will contact the authors to obtain more information if necessary. The grey literature and unpublished researches will not be searched. The MEDLINE search strategy is provided in table 1.

Table 1 Search String

MEDLINE
"flow mediated vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) dependent vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) function (or dysfunction), or FMD or vascular reactivity, and brachial artery" AND "cardiovascular diseases (or coronary disease) or masked hypertension (or hypertension or white coat hypertension or pre-hypertension)"

Eligibility criteria

Population: the focus population are adults (over 18 years old or older without upper age limit). Persons maybe have or not have elevated BP or from hypertensive groups (systolic BP $\geq$ 140 mmHg or diastolic BP $\geq$ 90 mmHg or take antihypertensive drugs for hypertension). Blood pressure is quantified by each method in studies, including office BP, HBPM, ABPM, or can be acquainted by published results or illustrated by authors.

Exposure: any measurement of FMD utilizing any general metric. Instances where the FMD M $\pm$ SD data are recorded for more than two groups, data will be selected with the highest FMD versus the lowest FMD group and ensure that only one effect size is analyzed each study and that dissimilar n=k groups are pooled in analyses.

Outcomes: FMD, masked hypertension.

Study design

Experimental studies designed as randomized controlled trials and non-randomized trials, observational studies designed as a case-control study and longitudinal cohort will be eligible. Prospective and retrospective studies will also be eligible. We will exclude the studies of cross-sectional studies, case reports and series.

Exclusion

Studies utilizing only patient self-reported to determine masked hypertension including self-reported of a physician's diagnosis which is not further checked

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consistently with the definitions listed under study outcomes are excluded.

Study selection process

All citations identified through our search strategy will be imported into an EndNote library and eliminate duplicate data. First, two review authors will select the title and abstracts of studies by the search and identify the potentially eligible studies independently. Then they will check against the inclusion criteria. Any disagreement at all stages of the selection process will be solved through consultation or consensus with the corresponding author. A PRISMA flow chart will outline the study selection process and reasons for exclusions(shown in figure1).

Data items for collection

After the determination of the initial selected study, eligibility information will be extracted for each study relating to study identification (authors, year of publication, country where studies took place and other detailed information), study design and study characteristics (e.g. sample size, adverse events, duration of follow-up), characteristics of the population under study (age, sex, education, body mass index, hypercholesterolemia, diabetes, kidney disease, liver disease, stroke, coronary heart disease, and heart failure), methods of measuring BP. Primary outcome data collected will include office-BP(systolic and diastolic blood pressure), 24-hour ABPM(systolic and diastolic blood pressure), hypertension phenotypes, brachial artery FMD outcomes, reported either as categorical numbers (numerator and denominator). These data will be extracted by one writer for all studies, and then the extracted data will be checked by a second writer to reduce individual errors and bias. All disagreements will be resolved by agreement of the two writers.

Risk of bias

We will utilize the RTI item bank to identify methodological bias among the included studies at the study-level (supplementary appendix 2)[15].

Patient and public involvement

No patients and members of the public will be directly involved. Only data already existent in the literature and the aforementioned sources will be used for this study.

Synthesis of data and summary measures

Data synthesis

We will provide a detailed description of the study results in both text and table lists for all included studies. We will describe these studies qualitatively relevant to study identifications (authors, year of publication, country where studies took place and other detailed information), study design and study characteristics (experimental or observational, sample size, adverse events, duration of follow-up), patients (age, sex, BMI), the methods used to measure different blood pressure and the value of FMD.

Meta-analysis

We will use Comprehensive Meta-Analysis Version 2.0 to conduct the meta-analyses. The summary effect measures may include d family effect sizes (e.g., hazard ratios, relative risk, odds ratios, Cohen's d) or r family effect sizes (e.g., r, β). When data are available to be pooled together, we will use a random-effects model using the inverse variance method which provides a more conservative estimate of effect size. In studies where the effect size is not reported, we will extract the individual cell data

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and calculate the risk ratio and 95% CI where possible. The r family effect sizes will be converted to the common metric r and pooled together with their associated 95% CI.

Data heterogeneity will use two measures of between-study heterogeneity: the I^2 statistic and Cochran Q-statistic. According to the Cochrane Handbook for Systematic Reviews[16 17], I^2 of 0–60% can be regarded as moderate unimportant(0–60%), while $I^2 > 60\%$ indicates substantial heterogeneity[18].

Planned subgroup and sensitivity analyses

Subgroup analyses will be conducted for the combined masked hypertension outcome stratified by sex and age if it is possible. In case that stratified sex or age analyses is not able to process, we will use meta-regression to conduct analyses adjusted for the percentage of males and females in the total sample, and mean or median of age. Sensitivity analyses will perform the effects of age < 50 and > 50 years old, or as mean or median age in meta-regression alternatively. This is based on the prevalence of hypertension increased with increasing age[19].

Assessment of publication bias

Egger's regression asymmetry method will be used to evaluate the publication bias[20].

GRADE framework for quality of evidence

The included study will use the Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidelines[21] to determine the quality of evidence and the strength of recommendations.

Discussion

The purpose of this study is intended to add to the literature by aggregating data concerning the risk of masked hypertension and endothelial dysfunction measured by FMD. Our study will contribute to the literature by clarifying whether FMD, derived from ultrasound imaging, is associated with these blood pressure outcomes. This is truly the consistent evidence that there is an increased risk of cardiovascular disease (CVD) associated with masked hypertension[22]. The findings might also potentially inform evidence-based clinical practice and policy regarding blood pressure measurement and hypertension management, especially in the diagnosis of pre-hypertension and masked hypertension.

There are a few limitations that will contextualize the findings and application of the study, because that endothelial dysfunction is the multiple results of complex interactions between cardiovascular regulatory mechanisms and environmental and behavioral reasons[23]. As such, we may not be able to identify antecedent risk factors for higher FMD or ways to modulate in order to lessen masked hypertension risk. Similarly, FMD can be dependent on blood pressure in some methodologies; thus, it may be difficult to extrapolate the effects of FMD independent from blood pressure. The included studies will also probably measure blood pressure at different methods, which could cause methodological heterogeneity. Limitations will also associate with the outcome of masked hypertension with different levels of effectiveness and heterogeneity. The ability to correctly judge the prognosis of masked hypertension outcomes will be related to the age, follow-up time and disease of the participants. This limitation is considerable in our proposed study's background since we are

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assessing the correlation between FMD and masked hypertension. Limitations of the original studies may also include between study heterogeneity and high risk of bias that will potentially limit the conclusions drawn. Moreover, as the included studies will include only English and Chinese language, the generalizability of the studies results published in other languages and other healthcare settings is finite measurement.

In conclusion, given there is no study available on the assessment of endothelial function measured by FMD in subject with masked-hypertension, there is a pressing demand to perform a meta-analysis on the performances of FMD for diagnosis of masked hypertension. The study will help to summarize the available research evidence, and the findings may have guiding significance for clinical practice and diagnosis of hypertension.

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References

1. Judd E, Calhoun DA. Apparent and true resistant hypertension: definition, prevalence and outcomes. *J Hum Hypertens* 2014;28(8):463-8. doi: 10.1038/jhh.2013.140 [published Online First: 2014/01/17]

2. Anstey DE, Pugliese D, Abdalla M, et al. An Update on Masked Hypertension. *Current hypertension reports* 2017;19(12):94. doi: 10.1007/s11906-017-0792-4 [published Online First: 2017/10/27]

3. Franklin SS, O'Brien E, Staessen JA. Masked hypertension: understanding its complexity. *European heart journal* 2017;38(15):1112-18. doi: 10.1093/eurheartj/ehw502 [published Online First: 2016/11/12]

4. Franklin SS, Thijs L, Li Y, et al. Masked hypertension in diabetes mellitus: treatment implications for clinical practice. *Hypertension (Dallas, Tex : 1979)* 2013;61(5):964-71. doi: 10.1161/hypertensionaha.111.00289 [published Online First: 2013/03/13]

5. Gaborieau V, Delarche N, Gosse P. Ambulatory blood pressure monitoring versus self-measurement of blood pressure at home: correlation with target organ damage.

Journal of hypertension 2008;26(10):1919-27. doi: 10.1097/HJH.0b013e32830c4368 [published Online First: 2008/09/23]

6. Conen D, Bamberg F. Noninvasive 24-h ambulatory blood pressure and cardiovascular disease: a systematic review and meta-analysis. *Journal of hypertension* 2008;26(7):1290-9. doi: 10.1097/HJH.0b013e3282f97854 [published Online First: 2008/06/14]

7. Zhang L, Li Y, Wei FF, et al. Strategies for classifying patients based on office, home, and ambulatory blood pressure measurement. *Hypertension (Dallas, Tex : 1979)* 2015;65(6):1258-65. doi: 10.1161/hypertensionaha.114.05038 [published Online First: 2015/04/15]

8. Stergiou GS, Salgami EV, Tzamouranis DG, et al. Masked hypertension assessed by ambulatory blood pressure versus home blood pressure monitoring: is it the same phenomenon? *American journal of hypertension* 2005;18(6):772-8. doi: 10.1016/j.amjhyper.2005.01.003 [published Online First: 2005/06/01]

9. Celermajer DS, Sorensen KE, Gooch VM, et al. Non-invasive detection of endothelial dysfunction in children and adults at risk of atherosclerosis. *Lancet (London, England)* 1992;340(8828):1111-5. doi: 10.1016/0140-6736(92)93147-f [published Online First: 1992/11/07]

10. Korkmaz H, Onalan O. Evaluation of endothelial dysfunction: flow-mediated dilation. *Endothelium : journal of endothelial cell research* 2008;15(4):157-63. doi: 10.1080/10623320802228872 [published Online First: 2008/07/30]

11. Inoue T, Matsuoka H, Higashi Y, et al. Flow-mediated vasodilation as a diagnostic

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modality for vascular failure. *Hypertension research : official journal of the Japanese Society of Hypertension* 2008;31(12):2105-13. doi: 10.1291/hypres.31.2105 [published Online First: 2009/01/14]

12. Frick M, Suessenbacher A, Alber HF, et al. Prognostic value of brachial artery endothelial function and wall thickness. *Journal of the American College of Cardiology* 2005;46(6):1006-10. doi: 10.1016/j.jacc.2005.05.070 [published Online First: 2005/09/20]

13. Fathi R, Haluska B, Isbel N, et al. The relative importance of vascular structure and function in predicting cardiovascular events. *Journal of the American College of Cardiology* 2004;43(4):616-23. doi: 10.1016/j.jacc.2003.09.042 [published Online First: 2004/02/21]

14. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS medicine* 2009;6(7):e1000100. doi: 10.1371/journal.pmed.1000100 [published Online First: 2009/07/22]

15. Viswanathan M, Berkman ND, Dryden DM, et al. AHRQ Methods for Effective Health Care. Assessing Risk of Bias and Confounding in Observational Studies of Interventions or Exposures: Further Development of the RTI Item Bank. Rockville (MD): Agency for Healthcare Research and Quality (US) 2013.

16. Moreira MT, Smith LA, Foxcroft D. Social norms interventions to reduce alcohol misuse in university or college students. *The Cochrane database of systematic reviews* 2009(3):Cd006748. doi: 10.1002/14651858.CD006748.pub2 [published Online First:

For peer review only

2009/07/10]

17. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in medicine* 2002;21(11):1539-58. doi: 10.1002/sim.1186 [published Online First: 2002/07/12]

18. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ (Clinical research ed)* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

19. Takase H, Tanaka T, Takayama S, et al. Recent changes in blood pressure levels, hypertension prevalence and treatment rates, and the rate of reaching target blood pressure in the elderly. *Medicine* 2017;96(50):e9116. doi: 10.1097/md.0000000000009116 [published Online First: 2018/02/03]

20. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ (Clinical research ed)* 1997;315(7109):629-34. doi: 10.1136/bmj.315.7109.629 [published Online First: 1997/10/06]

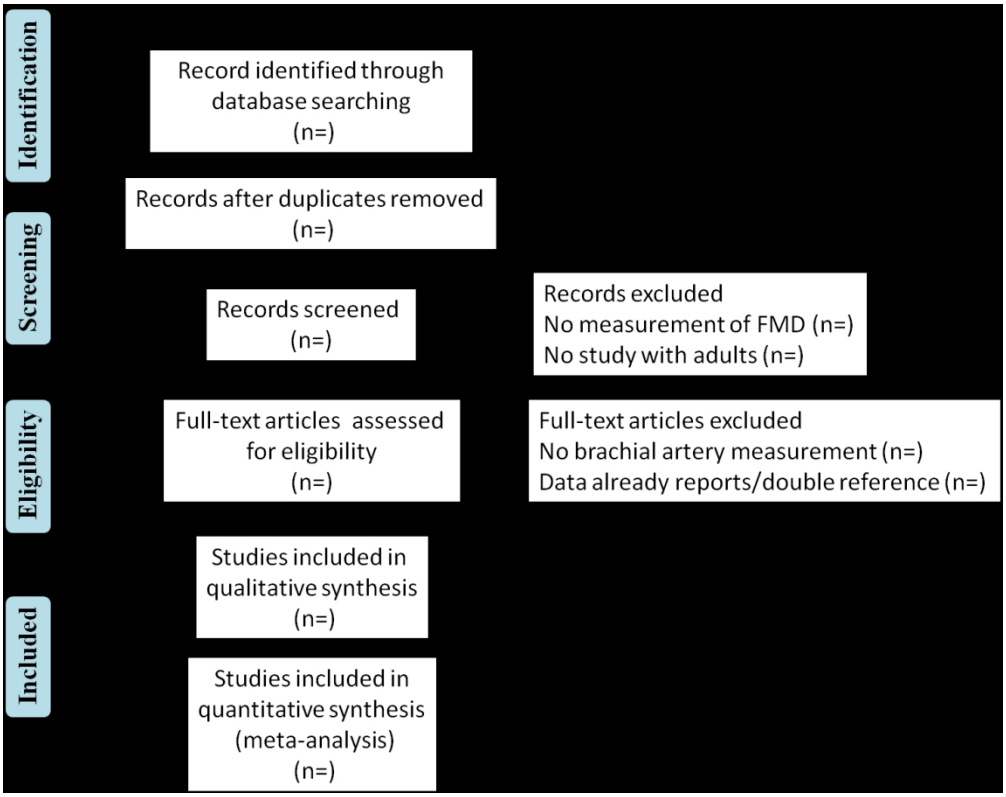
21. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ (Clinical research ed)* 2008;336(7650):924-6. doi: 10.1136/bmj.39489.470347.AD [published Online First: 2008/04/26]

22. Pierdomenico SD, Cuccurullo F. Prognostic value of white-coat and masked hypertension diagnosed by ambulatory monitoring in initially untreated subjects: an updated meta analysis. *American journal of hypertension* 2011;24(1):52-8. doi: 10.1038/ajh.2010.203 [published Online First: 2010/09/18]

For peer review only

23. Charakida M, Masi S, Lüscher TF, et al. Assessment of atherosclerosis: the role of flow-mediated dilatation. *European heart journal* 2010;31(23):2854-61. doi: 10.1093/eurheartj/ehq340 [published Online First: 2010/09/25]

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248x195mm (150 x 150 DPI)

Appendix 1

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review		√	
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		√	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	√		32
Authors:					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	√		
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	√		
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		√	
Support:					
Sources	5a	Indicate sources of financial or other support for the review		√	
Sponsor	5b	Provide name for the review funder and/or sponsor		√	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		√	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	√		46-89
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	√		107-117
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	√		119-122
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	√		98-104
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	√		Table 1
Study records:					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	√		164-176
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	√		128-131
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	√		131-134
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	√		136-142
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	√		142-147
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the	√		150-151

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		outcome or study level, or both; state how this information will be used in data synthesis			
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	√		158-162
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	√		173-176
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	√		178-184
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned		√	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	√		186
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	√		188-190

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Appendix 2

RTI Risk of Bias Item Bank

Risk of bias item	Description	
1. **Retrospective/Prospective**	**Is the study design prospective, retrospective, or mixed?** *[Abstractor: Prospective design requires that the outcome has not occurred at the time the study is initiated and information is collected over time to assess relationships with the outcome (and includes nested case-control studies). Mixed design includes case-control or cohort studies in which one group is studied prospectively and the other retrospectively. A retrospective design analyzes data from past records. The question is not applicable to cross-sectional studies.]*	**Prospective..... Mixed..... Retrospective..... Cannot determine/not applicable..... Explanation for rating:**
2. **Inclusion/Exclusion Criteria**	**Are critical inclusion/exclusion criteria clearly stated (does not require the reader to infer)?** *[Principal Investigator (PI): Provide direction to abstractors by listing individual criteria of a priori significance and minimal requirements for criteria to be considered "clearly stated." Include this question to identify specific inclusion/exclusion criteria that should be consistently recorded across studies] [Abstractor: Use "Partially" if only some criteria are stated or if some criteria are not clearly stated (corresponding to directions provided by the PI). Note that studies may describe inclusion criteria alone (i.e., include x), exclusion criteria (i.e., do not include x), or a combination of inclusion and exclusion criteria.]*	**Yes..... Partially: some, but not all, criteria stated or some criteria not clearly stated..... No..... Explanation for rating:**
3. **Inclusion/Exclusion Criteria**	**Are the inclusion/exclusion criteria measured using valid and reliable measures?** *[PI: Separately specify each criterion that abstractors should consider based on its relevance to study bias. It is unlikely that all criteria will need to be evaluated in relation to this question. Provide direction to abstractors on valid and reliable measurement of each criterion that is to be considered. For example, prior exposure or disease status is a frequent inclusion/exclusion criterion, particularly in inception cohorts. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such*	**Yes..... No..... Cannot determine; measurement approach not reported..... Explanation for rating:**

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	as clinical reports and lab findings. Replicate question to evaluate each individual inclusion/exclusion criterion.]	
4. **Inclusion/Exclusion Criteria**	**Did the study apply inclusion/exclusion criteria uniformly to all comparison groups/arms of the study?** *[PI: Drop question if not relevant to entire body of evidence (e.g., all case-series, singlearm studies).]*	**Yes..... Partially: some, but not all criteria, applied to all arms or not clearly stated if some criteria are applied to all arms ..... No..... Cannot determine: article does not specify..... Not applicable: study has only one arm and so does not include comparison groups..... Explanation for rating:**
5. **Recruitment across groups**	**Was the strategy for recruiting participants into the study the same across study groups/arms of the study?** ***[PIs: This question is likely to be more relevant for prospective or mixed designs than retrospective designs. Drop question if not relevant to entire body of evidence (e.g., all studies generally have only one arm).]***	**Yes..... No..... Cannot determine ..... Not applicable: one study group/arm ..... Explanation for rating:**
6. **Statistical power**	**Was the sample size sufficiently large to detect a clinically significant difference of 5% or more between groups in at least one primary outcome measure?** *[PI: Specify a different percent, if clinically relevant for each outcome of interest. Question relates to precision; reviewers whose evaluation of quality is limited to considerations of systematic error or risk of bias (not random error/precision) need not include this question. Reviewers who include both precision and systematic error in their evaluation of quality but rely on meta-analysis for pooled estimates need not include this question. PIs who choose to include considerations of precision in their assessment may include the question, but should be aware of the need for collaboration between clinical and statistical expertise in determining the threshold for a clinically adequate sample size.]*	**Yes..... No..... Explanation for rating:**
7.	***What is the level of detail in describing***	**High: very clear, all PI-**

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Detail of exposure	***the intervention or exposure?*** *[PI: Specify which details need to be stated (e.g., intensity, duration, frequency, route, setting, and timing of intervention/exposure). For case-control studies, consider whether the condition, timing, frequency, and setting of symptoms are provided in the case definition. PI needs to establish criteria for high, medium, or low response.]*	**required details provided** **Medium: somewhat clear, majority of PI required details provided** **.....** **Low: unclear, many PI-required details missing** **.....** **Explanation for rating:**
8. **Specification of outcomes**	**Are the important outcomes prespecified by the researchers? Do not consider harms in answering this question unless they should have been pre-specified.** *[PI: This question can be asked for all outcomes together or replicated for each event. Each adverse event of interest should be specified for abstractors. Relevant source information includes all study data, including what may have been established in relation to an initial randomized controlled trial. Drop question if not relevant (e.g., primary outcome for case-control studies).]*	**Yes** **.....** **Partially** **.....** **No** **.....** **Not applicable** **.....** **Explanation for rating:**
9. **Appropriate comparison group**	**Is the selection of the comparison group appropriate, after taking into account feasibility and ethical considerations.** *[PI: Provide instruction to the abstractor based on the type of study. Interventions with community components are likely to have contamination if all groups are drawn from the same community. Interventions without community components should select groups from the same source (e.g., community or hospital) to reduce baseline differences across groups. For case-control studies, controls should represent the population from which cases arose; that is, controls should have met the case definition if they had the outcome.]*	**Yes.....** **No.....** **Cannot determine or no description of the derivation of the comparison group** **.....** **Not applicable: study does not include a comparison group (case series, one study arm) .....** **Explanation for rating:**
10. **Attempt To Balance**	**Any attempt to balance the allocation between the groups (e.g., through stratification, matching, propensity scores).** *[PI: This is most likely to be used in case-control study designs. Drop if not relevant to the body of evidence.]*	**Yes or study accounts for imbalance between groups through a post hoc approach such as multivariate analysis** **.....** **No or cannot determine** **.....** **Not applicable: study does not include a comparison group (case**

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		series or one study arm) **Explanation for rating:**
11. **Contamination**	**Did researchers isolate the impact from a concurrent intervention or an unintended exposure that might bias results, e.g., through multivariate analysis, stratification, or subgroup analysis?** *[PI: specify interventions or exposures for abstractors.]*	**Yes.....** **Partially.....** **.** **No or do not know: concurrent intervention or Unintended exposure is not described)** **.....** **Not applicable: no concurrent interventions or unintended exposures likely.....** **Explanation for rating:**
12. **Variation in execution of protocol**	**Did execution of the study vary from the intervention protocol proposed by the investigators and therefore compromise the conclusions of the study?** *[PI: Consider intensity, duration, frequency, route, setting, and timing of intervention/exposures. Drop if not relevant for body of literature.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Not applicable: not an intervention study** **.....** **Explanation for rating:**
13. **Blind Outcomes Assessment**	**Were the outcome assessors blinded to the intervention or exposure status of participants?** *[PI: There may be circumstances where clinical evaluators cannot be blinded to exposure status. Drop if not relevant to the body of literature.]*	**Yes.....** **No.....** **Not applicable: assessor cannot be blinded.....** **Explanation for rating:**
14. **Exposures assessed using valid**	**Are interventions/exposures assessed using valid and reliable measures, implemented consistently across all study participants?** *[PI: Important measures may be listed separately. PI may need to establish a threshold for what would constitute acceptable measures based on study topic. When subjective or objective measures could be collected, subjective measures based on selfreport may be considered as being less reliable and valid than objective measures such as clinical reports and lab findings. Replicate question when needed.]*	**Yes.....** **No.....** **Cannot determine or measurement approach not reported** **.....** **Explanation for rating:**
15. **Source of Information for Outcomes**	**Are outcomes assessed using valid and reliable measures, implemented consistently across all study participants?**	**Yes.....** **No.....** **Cannot determine or measurement approach**

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	[PI: Primary outcomes should be identified for abstractors and if there is more than one, they may be listed separately. Also, identify any relevant secondary outcomes and harms. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such as clinical reports and lab findings. Note for case-control studies: consider whether the ascertainment of cases was independent of exposure.]	**not reported** **.....** **Explanation for rating:**
16. **Equality of Length of Follow-Up for Participants**	**Is the length of follow-up the same for all groups?** *[For case-control studies, are cases and controls matched on length of followup? Abstractor: When follow-up was the same for all study participants, the answer is yes. If different lengths of follow-up were adjusted by statistical techniques, (e.g., survival analysis), the answer is yes. Studies in which differences in follow-up were ignored should be answered no.]*	**Yes.....** **No or cannot determine** **.....** **Not0 applicable: cross-sectional or only one group followed over time.....** **Explanation for rating:**
17. **Length of Follow-up Adequate**	**Is the length of time following the intervention/exposure sufficient to support the evaluation of primary outcomes and harms?** *[PI: Primary outcomes (including harms) should be identified for abstractors. Important measures may be listed separately. Abstractors should be provided with specific criteria for sufficient length of follow-up based on prior research or theory. Drop if entire body of evidence is cross-sectional or if minimal length of follow-up period is specified through inclusion criteria.]*	**Yes.....** **Partially: some primary outcomes are followed for a sufficient length of time.....** **No.....** **Cannot determine** **.....** **Not applicable: cross-sectional** **.....** **Explanation for rating:**
18. **Completeness of Follow-Up**	**Did attrition from any group exceed [x] percent?** *[PI: Attrition is measured in relation to the time between baseline (allocation in some instances) and outcome measurement for both retrospective and prospective studies and could include data loss from crossover. Attrition rates may vary by outcome and time of measurement. Specify the criterion to meet relevant standards for the topic. Specify measurement period of interest, if repeated measures. Cochrane standard for attrition is 20 percent for shorter term (<1 year) and 30 percent for longer term (≥ 1 year). Drop if entire body of evidence is cross-sectional]*	**Yes.....** **No.....** **Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional.....** **Explanation for rating:**
19. **Attrition difference**	**Did attrition differ between groups by more than 20 percent?**	**Yes.....** **No.....**

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	[PI: If appropriate, modify difference criterion to meet relevant standards for the topic. Attrition rates may vary by outcome and time of measurement. Drop if entire body of evidence is cross-sectional or case series.]	**Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional or only one group followed—case series, one-arm study.....** **Explanation for rating:**
20. **Assessment of Baseline Comparability**	**Does the analysis control for baseline differences between groups?** *[PI: Drop if entire body of evidence is case series or case control. Define adequate control. List critical baseline differences that need to be controlled.]*	**Yes.....** **No.....** **Insufficient reporting to be able to determine .....** **Not applicable: only one group, no comparison group (case series), or case control study, no difference in measured baseline characteristics .....** **Explanation for rating:**
21. **Identification of Prognostic Factors (Effect Modifiers and Confounders)**	**Are confounding and/or effect modifying variables assessed using valid and reliable measures across all study participants?** *[PI: Some characteristics may require that sources for establishing their validity and/or reliability be described or referenced. If so, provide instruction to abstractors.]*	**Yes.....** **No.....** **Cannot determine or source for measures not reported.....** **Not applicable: no confounders or effect modifiers included in the study.....** **Explanation for rating:**
22. **Case-Mix Adjustment**	**Were the important confounding and effect modifying variables taken into account in the design and/or analysis (e.g., through matching, stratification, interaction terms, multivariate analysis, or other statistical adjustment)?** *[PI: Provide instruction to abstractors on adequate adjustment for confounding and testing for effect modification.]*	**Yes.....** **Partially: some variables taken into account or adjustment achieved to some extent.....** **No: not accounted for or not identified.....** **Cannot determine .....** **Explanation for rating:**
23. **Intention-to-Treat Analysis**	**In cases of high loss to follow-up (or differential loss to follow-up), is the impact assessed (e.g., through sensitivity**	**Yes.....** **No.....** **Cannot determine**

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	analysis or other adjustment method)?	**.....** **Not applicable: no loss to follow-up or loss to follow-up was not considered to be high, crosssectional study, or case-control study selected on outcome** **.....** **Explanation for rating:**
24. **Primary outcomes missing**	**Are any important primary outcomes missing from the results?** *[PI: Identify all primary outcomes, including timing of measurement, that one would expect to be reported in the study.]*	**Yes.....** **No.....** **Cannot determine** **.....** **Explanation for rating:**
27. **Appropriate statistics for outcome**	***Are the statistical methods used to assess the main harm or adverse event outcomes appropriate to the data?*** *[Abstractor: Question relates to precision and may not be relevant for systematic reviews that are able to pool data. The statistical techniques used must be appropriate to the data and take into account issues such as controlling for dose-response, small sample size, clustering, rare outcomes, and multiple comparisons. In normally distributed data, the standard error, standard deviation, or confidence intervals should be reported. In non-normally distributed data, inter-quartile range should be reported.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Explanation for rating:**
28. **Appropriate interpretation**	**Are results believable taking study limitations into consideration?** *[Abstractor: This question is intended to capture the overall quality of the study. Consider issues that may limit your ability to interpret the results of the study. Review responses to earlier questions for specific criteria.]*	**Yes.....** **Partially.....** **.** **No.....** **Explanation for rating:**
29. **Funding**	**Is the source of funding identified?** *[PI: The relevance of this question will depend upon the topic. This question may be modified to identify particular sources of funding (e.g., industry, government, university, or foundation funding).]*	**Yes.....** **No.....** **Explanation for rating:**

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Contributors: ZX conceived the study and drafted the protocol. LY and GL revised it. ZX, GY, YQ, ZC and ZW developed the search strategies and will run it. ZX and will select studies and extract data. ZX and ZC will analyze the data. All authors have approved the final edition of the protocol.

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Correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation：a protocol of systematic review and meta-analysis

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Abstract

Introduction

A surrogate marker to evaluate artery endothelial response when stimulated by reactive hyperemia, known as brachial flow-mediated dilation (FMD), has prognostic value for predicting hypertensive organ damage and cardiovascular disease (CVD) events. However, the degree of correlation between brachial artery FMD and masked hypertension outcomes remains less clear. This study's purpose is to pool data regarding FMD with respect to masked hypertension.

Methods and analysis

Electronic databases, MEDLINE, EMBASE, China National Knowledge Infrastructure (CNKI) and the Cochrane Library, will be searched for key words of masked hypertension, treated but uncontrolled hypertension, isolated ambulatory hypertension, and white coat hypertension. Authors and reference inventory for included studies also will be contacted to identify additional published and unpublished studies. Eligibility criteria are as shown: population—adults (18 years old or older with no upper limit) as a baseline, from hypertensive groups (systolic blood pressure ≥ 140 mmHg and/or diastolic BP ≥ 90 mmHg or take antihypertensive drugs for hypertension) and from primary care, community cohorts, electronic database registries, or randomized and controlled trials; exposures—any metrics for FMD from ambulatory BP monitoring, home BP monitoring, or office BP; comparisons—individuals without masked hypertension who do not have elevated

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BP; and outcomes—masked hypertension or masked uncontrolled hypertension. Two writers will be engaged in screening and collecting data independently, and disagreements resolved through discussion. Data extraction will include but not limited to basic data designated as hazard ratios, odds ratios, correlations, regression coefficients, and original cell data if available. We will use Comprehensive Meta-Analysis Version 2.0 to conduct the meta-analysis.

Ethics and dissemination

This study does not require ethics approval. It will be disseminated internationally and presented in peer-reviewed literature.

PROSPERO registration number: CRD42020208362

Strengths and limitations of this study

- This is the first meta-analysis we are aware of to report the correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation.
- The results provide the evidence of a new method to evaluate masked hypertension apart from Ambulatory Blood Pressure Monitoring.
- Since we include different types of studies, bias cannot be avoided to some extent.
- We distinguish masked uncontrolled hypertension from masked hypertension through conducting a subgroup analysis.

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Background

Masked hypertension (MH) is defined as possessing a normal blood pressure in the office despite having an elevated measurement outside of it[1]. Diagnosing masked hypertension (MH) still has inherent challenges. Both missed diagnoses and excess diagnoses have undesirable consequences. Undiagnosed MH could frequently develop into damage at targeted organs prior to transitioning to sustained hypertension[2]. Masked control of hypertension is more common in patients with hypertension, with a prevalence rate of 30-50%[3]. MH is used to describe as a 24-hour mean increase in ambulatory blood pressure monitoring ABPM (SBP ≥ 130 mmHg and/or diastolic ≥ 80 mmHg) in patients whose blood pressure is controlled below optimal levels according to ABPM, despite clinical control. Moreover, it was discovered that taking antihypertensive medication, which is defined by masked uncontrolled hypertension (MUCH), has an increased prevalence rate for masked hypertension as compared to those not receiving antihypertensive treatment[4]. A definitive and accurate diagnostic conclusion is therefore required for all patients with suspected MH. Although studies have shown that hypertension is associated with an increased risk of CVD, the actual concern surrounding this health risk is not the mild elevation in BP that may be measured in a clinic, but masked hypertension being underestimated.

The diagnostic strategy starts with a clinical probability assessment and measuring blood pressure (BP) in the clinic. In patients with a high/likely clinical probability, ABPM and home blood pressure monitoring (HBPM) are both required. ABPM is currently the predominant test for MH. The test has been universally verified in terms of diagnostic accuracy and outcome studies[5-7]. However, ABPM has some limitations, including inconvenient when compared with HBPM[8], difficulty to operate, and a continuous concern regarding possible over diagnosing and overtreating hypertension. In recent years, the technology around quantifying endothelial function has rapidly evolved, which allowed for the introduction of flow-mediated dilation, a non-invasive method of assessing endothelial dysfunction[9]. Flow-mediated dilation (FMD) has been reported to be a predictor for hypertensive organ damage and CVD events[10 11]. FMD has already largely replaced the invasive method administered during cardiac catheterization. Many studies aimed at assessing the correlation between FMD and CVD risk. More recently, a few systematic reviews and meta-analyses have been published, and concluded that FMD is positively associated with a risk for future cardiovascular events[12 13]. However, the clinical community remains uncertain about the relationship between MH or MUCH diagnosed through ABPM and flow-mediated vasodilation responses. Indeed, the concrete relationship that exists between FMD and MH or MUCH is still unclear. For example, in some studies, there is a bias inherent in FMD results, thus influencing the ultimate diagnosis conclusion. Therefore, although previously published meta-analyses focusing on brachial FMD have been reported for CVD outcomes, summarized result related to MH or MUCH is still lacked.

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Methods

Aims

The study aims to synthesize the evidence regarding FMD and masked hypertension. The reporting of this study protocol conforms to the PRISMA-P guidelines (supplementary appendix 1) and was registered on the PROSPERO database[14]. The full study will adhere to all PRISMA guidelines. Any updates to this study will be registered on PROSPERO.

Search strategy

We will search for relevant articles in English and Chinese in these databases: MEDLINE, EMBASE, CNKI and the Cochrane Library. The search period will be from the date when each database was established to November 2020. We will perform a manual search of the reference lists of articles selected to supplement the electronic search. We will contact the authors to obtain more information if necessary. Gray literature and unpublished research will be excluded. The MEDLINE search strategy is provided in Table 1.

Table 1 Search String

MEDLINE
"flow mediated vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) dependent vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) function (or dysfunction), or FMD or vascular reactivity, and brachial artery" AND "cardiovascular diseases (or coronary disease) or masked hypertension (or hypertension or white coat hypertension or pre-hypertension or masked uncontrolled hypertension)"

Eligibility criteria

Population: the focus population is adults (18 years old or older without an upper age limit); the definition of masked hypertension was explicit (in-office blood pressure is normal (BP<135/85 mm Hg) (controlled or not controlled), daytime or home blood pressure out of the office is > 135/85 mmHg, and the 24-hour blood pressure is greater than 130/80 mmHg) [15]

Exposure: any measurement of FMD utilizing any general metric. Instances where the FMD M ± SD data were recorded for more than two groups, data will be selected with the highest FMD and not the lowest FMD group to ensure that only one effect size is analyzed for each study and that dissimilar n=k groups are pooled in analyses.

Outcomes: FMD, masked hypertension.

Study design

Experimental studies designed as randomized controlled trials and non-randomized trials, observational studies designed as case-control studies and longitudinal cohort will be eligible. Prospective and retrospective studies will also be eligible. We will exclude any cross-sectional studies, case reports, and series.

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Exclusion

Studies utilizing only patient self-reporting to determine masked hypertension, including self-reporting in a physician's diagnosis, which is not further verified consistently with the definitions listed under the study outcomes are excluded.

Study selection process

All citations identified through our search strategy will be imported into an EndNote library and any duplicates will be eliminated. First, two review authors will select the titles and abstracts from studies by searching and identify potentially eligible studies independently. Then they will check against the inclusion criteria. Any disagreements at all stages during the selection process will be resolved through consultation or consensus with the corresponding author. A PRISMA flow chart will outline the study selection process and the reasons for any exclusions (as shown in Figure1).

Data items for collection

After determining the initial selection of studies, eligibility information will be extracted for each study relating to study identification (authors, years of publication, countries where the studies took place, and other information), study design and study characteristics (e.g. sample size, adverse events, duration of follow-ups), characteristics of the populations being studies (age, sex, education, body mass index, hypercholesterolemia, diabetes, kidney disease, liver disease, stroke history, coronary heart disease, and heart failure), and methods of measuring BP. Any primary outcome data that is collected will include office-BP (systolic and diastolic blood pressure), 24-hour ABPM (systolic and diastolic blood pressure), and hypertension phenotypes, brachial artery FMD outcomes, reported as categorical numbers. This data will be extracted by one writer for all studies, and then the extracted data will be checked by a second writer to reduce individual errors and bias. All disagreements will be resolved between the two writers.

Risk of bias

We will utilize the RTI item bank to identify methodological bias among the included studies at the study-level (supplementary appendix 2)[16].

Patient and public involvement

No patients and members of the public will be directly involved. Only data that already exists in the literature and the aforementioned sources will be used for this study.

Synthesis of data and summary measures

Data synthesis

We will provide a detailed description of the study results for all included studies in both text and table lists. We will describe these studies that are qualitatively relevant to study identifications (authors, years of publication, countries where the studies took

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place and other information), study designs, and study characteristics (experimental or observational, sample size, adverse events, duration of follow-ups), patients (age, sex, BMI), the methods used to measure different blood pressures, and the values of FMD.

Meta-analysis

We will use Comprehensive Meta-Analysis Version 2.0 to conduct the meta-analyses. The summary effect measures may include d family effect sizes (e.g., hazard ratios, relative risk, odds ratios, Cohen's d) or r family effect sizes (e.g., r, β). When data is available to be pooled together, we will use a random-effects model using the inverse variance method, which provides a more conservative estimate of effect size. In studies where the effect size is not reported, we will extract the individual cell data and calculate the risk ratio and 95% CI where possible. The r family effect sizes will be converted into the common metric r and pooled together with their associated 95% CI.

Data heterogeneity will utilize two measures of between-study heterogeneity: the I^2 statistic and Cochran Q-statistic. According to the Cochrane Handbook for Systematic Reviews[17 18], I^2 of 0–60% can be regarded as moderately unimportant (0–60%), while $I^2 > 60\%$ indicates substantial heterogeneity[19].

Planned subgroup and sensitivity analyses

Considering the possibility of using different covariates in the study, we will conduct a subgroup analysis to assess the impact of each study feature on the study's primary outcome. We will conduct a subgroup analysis of patients with MH and MUCH. Also, subgroup analyses will be conducted for combined masked hypertension outcomes stratified by sex and age if possible. In case that stratified sex or age analyses are not able to be processed, we will use meta-regression to conduct analyses adjusted for the percentage of males and females in the total sample and mean or median of age. Sensitivity analyses will conduct the effects of age < 50 and > 50 years old, or as mean or median age in meta-regression alternatively. This is based on the prevalence of hypertension increasing with age[20].

Assessment of publication bias

Egger's regression asymmetry method will be used to evaluate the publication bias[21].

GRADE framework for quality of evidence

The included study will use the Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidelines[22] to determine the quality of evidence and strength of recommendations.

Discussion

The purpose of this study is intended to add to the literature by aggregating data concerning the risk of masked hypertension and endothelial dysfunction measured by FMD. Our study will contribute to the literature by clarifying whether FMD, derived from ultrasound imaging, is associated with these blood pressure outcomes. This is

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the actual, consistent evidence that there is an increased risk of cardiovascular disease (CVD) associated with masked hypertension[23]. The findings might also potentially inform evidence-based clinical practices and policies regarding blood pressure measurement and hypertension management, especially when diagnosing pre-hypertension and masked hypertension.

There are a few limitations that will contextualize the findings and applications of the study because endothelial dysfunction is the result of multiple complex interactions between cardiovascular regulatory mechanisms and environmental and behavioral reasons[24]. As such, we may not be able to identify antecedent risk factors for higher FMD or ways to modulate and lessen masked hypertension risks. Similarly, FMD can depend on blood pressure in some methodologies; thus, it may be difficult to extrapolate the effects of FMD independent from blood pressure. The included studies will also probably measure blood pressure by different methods, which could cause methodological heterogeneity. Limitations will also be associated with the outcomes of masked hypertension based on different levels of effectiveness and heterogeneity. The ability to correctly judge the prognosis of masked hypertension outcomes will be related to the ages, follow-up durations, and diseases of the participants. This limitation is a serious concern for our proposed study's background since we are assessing the correlation between FMD and masked hypertension. Limitations in the original studies may also include study heterogeneity and high risk of bias, which will potentially limit the conclusions drawn. Moreover, as the included studies will only be drawn from the English and Chinese languages, the generalizability of the study results published in other languages and other healthcare settings is finite in measurement.

In conclusion, given there are no studies available that assess endothelial functionality measured by FMD within the scope of masked-hypertension, and there is a pressing need to perform a meta-analysis on how FMD performs in terms of diagnosing masked hypertension. The study will help summarize the available research evidence, and the findings may have guiding significance for clinical practices and diagnosing hypertension.

Contributors: ZW proposed the conception and acquired funding. LY and ZC involved in the study design and project administration. ZX, GL, and ZC drafted the manuscript.

Competing interests: None declared.

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References

1. Anstey DE, Pugliese D, Abdalla M, et al. An Update on Masked Hypertension. *Current hypertension reports* 2017;19(12):94. doi: 10.1007/s11906-017-0792-4 [published Online First: 2017/10/27]

2. Franklin SS, O'Brien E, Staessen JA. Masked Hypertension: Understanding Its Complexity. *European heart journal* 2017;38(15):1112-18. doi: 10.1093/eurheartj/ehw502 [published Online First: 2016/11/12]

3. Banegas JR, Ruilope LM, de la Sierra A, et al. High Prevalence of Masked Uncontrolled Hypertension in People with Treated Hypertension. *Eur Heart J* 2014;35(46):3304-12. doi: 10.1093/eurheartj/ehu016 [published Online First: 2014/02/06]

4. Franklin SS, Thijs L, Li Y, et al. Masked Hypertension in Diabetes Mellitus: Treatment Implications for Clinical Practice. *Hypertension (Dallas, Tex : 1979)* 2013;61(5):964-71. doi: 10.1161/hypertensionaha.111.00289 [published Online First: 2013/03/13]

5. Gaborieau V, Delarche N, Gosse P. Ambulatory Blood Pressure Monitoring Versus Self-measurement of Blood Pressure at Home: Correlation with Target Organ Damage. *Journal of Hypertension* 2008;26(10):1919-27. doi: 10.1097/HJH.0b013e32830c4368 [published Online First: 2008/09/23]

6. Conen D, Bamberg F. Noninvasive 24-h Ambulatory Blood Pressure and Cardiovascular Disease: a Systematic Review and Meta-analysis. *Journal of Hypertension* 2008;26(7):1290-9. doi: 10.1097/HJH.0b013e3282f97854 [published Online First: 2008/06/14]

7. Zhang L, Li Y, Wei FF, et al. Strategies for Classifying Patients Based on Office, Home,

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and Ambulatory Blood Pressure Measurement. *Hypertension (Dallas, Tex : 1979)* 2015;65(6):1258-65. doi: 10.1161/hypertensionaha.114.05038 [published Online First: 2015/04/15]

8. Stergiou GS, Salgami EV, Tzamouranis DG, et al. Masked Hypertension Assessed by Ambulatory Blood Pressure Versus Home Blood Pressure Monitoring: is It the Same Phenomenon? *American Journal of Hypertension* 2005;18(6):772-8. doi: 10.1016/j.amjhyper.2005.01.003 [published Online First: 2005/06/01]

9. Celermajer DS, Sorensen KE, Gooch VM, et al. Non-invasive Detection of Endothelial Dysfunction in Children and Adults at Risk of Atherosclerosis. *Lancet (London, England)* 1992;340(8828):1111-5. doi: 10.1016/0140-6736(92)93147-f [published Online First: 1992/11/07]

10. Korkmaz H, Onalan O. Evaluation of Endothelial Dysfunction: Flow-mediated Dilation. *Endothelium : Journal of Endothelial Cell Research* 2008;15(4):157-63. doi: 10.1080/10623320802228872 [published Online First: 2008/07/30]

11. Inoue T, Matsuoka H, Higashi Y, et al. Flow-mediated Vasodilation As a Diagnostic Modality for Vascular Failure. *Hypertension Research : Official Journal of the Japanese Society of Hypertension* 2008;31(12):2105-13. doi: 10.1291/hypres.31.2105 [published Online First: 2009/01/14]

12. Frick M, Suessenbacher A, Alber HF, et al. Prognostic Value of Brachial Artery Endothelial Function and Wall Thickness. *Journal of the American College of Cardiology* 2005;46(6):1006-10. doi: 10.1016/j.jacc.2005.05.070 [published Online First: 2005/09/20]

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13. Fathi R, Haluska B, Isbel N, et al. The Relative Importance of Vascular Structure and Function in Predicting Cardiovascular Events. *Journal of the American College of Cardiology* 2004;43(4):616-23. doi: 10.1016/j.jacc.2003.09.042 [published Online First: 2004/02/21]

14. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA Statement for Reporting Systematic Reviews and Meta-analyses of Studies That Evaluate Health Care Interventions: Explanation and Elaboration. *PLoS Medicine* 2009;6(7):e1000100. doi: 10.1371/journal.pmed.1000100 [published Online First: 2009/07/22]

15. Rizzoni D. Masked Hypertension: How to Identify and When to Treat? *High Blood Press Cardiovasc Prev* 2016;23(3):181-6. doi: 10.1007/s40292-016-0140-9 [published Online First: 2016/04/05]

16. Viswanathan M, Berkman ND, Dryden DM, et al. AHRQ Methods for Effective Health Care. Assessing Risk of Bias and Confounding in Observational Studies of Interventions or Exposures: Further Development of the RTI Item Bank. Rockville (MD): Agency for Healthcare Research and Quality (US) 2013.

17. Moreira MT, Smith LA, Foxcroft D. Social Norms Interventions to Reduce Alcohol Misuse in University or College Students. *The Cochrane Database of Systematic Reviews* 2009(3):Cd006748. doi: 10.1002/14651858.CD006748.pub2 [published Online First: 2009/07/10]

18. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in medicine* 2002;21(11):1539-58. doi: 10.1002/sim.1186 [published Online First: 2002/07/12]

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19. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring Inconsistency in Meta-analyses. *BMJ (Clinical research ed)* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

20. Takase H, Tanaka T, Takayama S, et al. Recent Changes in Blood Pressure Levels, Hypertension Prevalence and Treatment Rates, and the Rate of Reaching Target Blood Pressure in the Elderly. *Medicine* 2017;96(50):e9116. doi: 10.1097/md.0000000000009116 [published Online First: 2018/02/03]

21. Egger M, Davey Smith G, Schneider M, et al. Bias in Meta-analysis Detected by a Simple, Graphical Test. *BMJ (Clinical research ed)* 1997;315(7109):629-34. doi: 10.1136/bmj.315.7109.629 [published Online First: 1997/10/06]

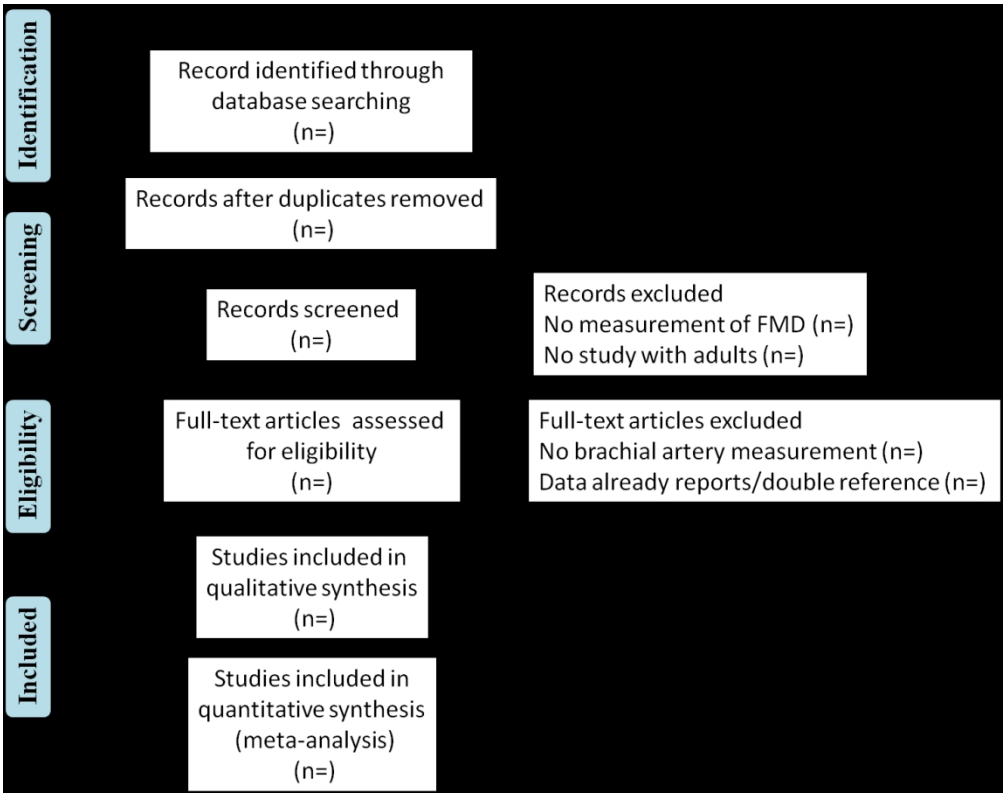
22. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: An Emerging Consensus on Rating Quality of Evidence and Strength of Recommendations. *BMJ (Clinical Research Ed)* 2008;336(7650):924-6. doi: 10.1136/bmj.39489.470347.AD [published Online First: 2008/04/26]

23. Pierdomenico SD, Cuccurullo F. Prognostic Value of White-coat and Masked Hypertension Diagnosed by Ambulatory Monitoring in Initially Untreated Subjects: An Updated Meta Analysis. *American Journal of Hypertension* 2011;24(1):52-8. doi: 10.1038/ajh.2010.203 [published Online First: 2010/09/18]

24. Charakida M, Masi S, Lüscher TF, et al. Assessment of Atherosclerosis: The Role of Flow-mediated Dilatation. *European Heart Journal* 2010;31(23):2854-61. doi: 10.1093/eurheartj/ehq340 [published Online First: 2010/09/25]

Figure 1 Flow chart of the study selection procedure

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248x195mm (150 x 150 DPI)

Appendix 1

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review		√	
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		√	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	√		32
Authors:					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	√		
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	√		
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		√	
Support:					
Sources	5a	Indicate sources of financial or other support for the review		√	
Sponsor	5b	Provide name for the review funder and/or sponsor		√	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		√	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	√		46-89
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	√		107-117
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	√		119-122
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	√		98-104
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	√		Table 1
Study records:					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	√		164-176
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	√		128-131
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	√		131-134
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	√		136-142
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	√		142-147
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the	√		150-151

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		outcome or study level, or both; state how this information will be used in data synthesis			
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	√		158-162
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	√		173-176
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	√		178-184
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned		√	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	√		186
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	√		188-190

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Appendix 2

RTI Risk of Bias Item Bank

Risk of bias item	Description	
1. **Retrospective/Prospective**	**Is the study design prospective, retrospective, or mixed?** *[Abstractor: Prospective design requires that the outcome has not occurred at the time the study is initiated and information is collected over time to assess relationships with the outcome (and includes nested case-control studies). Mixed design includes case-control or cohort studies in which one group is studied prospectively and the other retrospectively. A retrospective design analyzes data from past records. The question is not applicable to cross-sectional studies.]*	**Prospective..... Mixed..... Retrospective..... Cannot determine/not applicable..... Explanation for rating:**
2. **Inclusion/Exclusion Criteria**	**Are critical inclusion/exclusion criteria clearly stated (does not require the reader to infer)?** *[Principal Investigator (PI): Provide direction to abstractors by listing individual criteria of a priori significance and minimal requirements for criteria to be considered "clearly stated." Include this question to identify specific inclusion/exclusion criteria that should be consistently recorded across studies] [Abstractor: Use "Partially" if only some criteria are stated or if some criteria are not clearly stated (corresponding to directions provided by the PI). Note that studies may describe inclusion criteria alone (i.e., include x), exclusion criteria (i.e., do not include x), or a combination of inclusion and exclusion criteria.]*	**Yes..... Partially: some, but not all, criteria stated or some criteria not clearly stated..... No..... Explanation for rating:**
3. **Inclusion/Exclusion Criteria**	**Are the inclusion/exclusion criteria measured using valid and reliable measures?** *[PI: Separately specify each criterion that abstractors should consider based on its relevance to study bias. It is unlikely that all criteria will need to be evaluated in relation to this question. Provide direction to abstractors on valid and reliable measurement of each criterion that is to be considered. For example, prior exposure or disease status is a frequent inclusion/exclusion criterion, particularly in inception cohorts. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such*	**Yes..... No..... Cannot determine; measurement approach not reported..... Explanation for rating:**

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	as clinical reports and lab findings. Replicate question to evaluate each individual inclusion/exclusion criterion.]	
4. **Inclusion/Exclusion Criteria**	**Did the study apply inclusion/exclusion criteria uniformly to all comparison groups/arms of the study?** *[PI: Drop question if not relevant to entire body of evidence (e.g., all case-series, singlearm studies).]*	**Yes..... Partially: some, but not all criteria, applied to all arms or not clearly stated if some criteria are applied to all arms ..... No..... Cannot determine: article does not specify..... Not applicable: study has only one arm and so does not include comparison groups..... Explanation for rating:**
5. **Recruitment across groups**	**Was the strategy for recruiting participants into the study the same across study groups/arms of the study?** ***[PIs: This question is likely to be more relevant for prospective or mixed designs than retrospective designs. Drop question if not relevant to entire body of evidence (e.g., all studies generally have only one arm).]***	**Yes..... No..... Cannot determine ..... Not applicable: one study group/arm ..... Explanation for rating:**
6. **Statistical power**	**Was the sample size sufficiently large to detect a clinically significant difference of 5% or more between groups in at least one primary outcome measure?** *[PI: Specify a different percent, if clinically relevant for each outcome of interest. Question relates to precision; reviewers whose evaluation of quality is limited to considerations of systematic error or risk of bias (not random error/precision) need not include this question. Reviewers who include both precision and systematic error in their evaluation of quality but rely on meta-analysis for pooled estimates need not include this question. PIs who choose to include considerations of precision in their assessment may include the question, but should be aware of the need for collaboration between clinical and statistical expertise in determining the threshold for a clinically adequate sample size.]*	**Yes..... No..... Explanation for rating:**
7.	***What is the level of detail in describing***	**High: very clear, all PI-**

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Detail of exposure	***the intervention or exposure?*** *[PI: Specify which details need to be stated (e.g., intensity, duration, frequency, route, setting, and timing of intervention/exposure). For case-control studies, consider whether the condition, timing, frequency, and setting of symptoms are provided in the case definition. PI needs to establish criteria for high, medium, or low response.]*	**required details provided** **Medium: somewhat clear, majority of PI required details provided** **.....** **Low: unclear, many PI-required details missing** **.....** **Explanation for rating:**
8. **Specification of outcomes**	**Are the important outcomes prespecified by the researchers? Do not consider harms in answering this question unless they should have been pre-specified.** *[PI: This question can be asked for all outcomes together or replicated for each event. Each adverse event of interest should be specified for abstractors. Relevant source information includes all study data, including what may have been established in relation to an initial randomized controlled trial. Drop question if not relevant (e.g., primary outcome for case-control studies).]*	**Yes** **.....** **Partially** **.....** **No** **.....** **Not applicable** **.....** **Explanation for rating:**
9. **Appropriate comparison group**	**Is the selection of the comparison group appropriate, after taking into account feasibility and ethical considerations.** *[PI: Provide instruction to the abstractor based on the type of study. Interventions with community components are likely to have contamination if all groups are drawn from the same community. Interventions without community components should select groups from the same source (e.g., community or hospital) to reduce baseline differences across groups. For case-control studies, controls should represent the population from which cases arose; that is, controls should have met the case definition if they had the outcome.]*	**Yes.....** **No.....** **Cannot determine or no description of the derivation of the comparison group** **.....** **Not applicable: study does not include a comparison group (case series, one study arm) .....** **Explanation for rating:**
10. **Attempt To Balance**	**Any attempt to balance the allocation between the groups (e.g., through stratification, matching, propensity scores).** *[PI: This is most likely to be used in case-control study designs. Drop if not relevant to the body of evidence.]*	**Yes or study accounts for imbalance between groups through a post hoc approach such as multivariate analysis** **.....** **No or cannot determine** **.....** **Not applicable: study does not include a comparison group (case**

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		series or one study arm) **Explanation for rating:**
11. **Contamination**	**Did researchers isolate the impact from a concurrent intervention or an unintended exposure that might bias results, e.g., through multivariate analysis, stratification, or subgroup analysis?** *[PI: specify interventions or exposures for abstractors.]*	**Yes.....** **Partially.....** **.** **No or do not know: concurrent intervention or Unintended exposure is not described)** **.....** **Not applicable: no concurrent interventions or unintended exposures likely.....** **Explanation for rating:**
12. **Variation in execution of protocol**	**Did execution of the study vary from the intervention protocol proposed by the investigators and therefore compromise the conclusions of the study?** *[PI: Consider intensity, duration, frequency, route, setting, and timing of intervention/exposures. Drop if not relevant for body of literature.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Not applicable: not an intervention study** **.....** **Explanation for rating:**
13. **Blind Outcomes Assessment**	**Were the outcome assessors blinded to the intervention or exposure status of participants?** *[PI: There may be circumstances where clinical evaluators cannot be blinded to exposure status. Drop if not relevant to the body of literature.]*	**Yes.....** **No.....** **Not applicable: assessor cannot be blinded.....** **Explanation for rating:**
14. **Exposures assessed using valid**	**Are interventions/exposures assessed using valid and reliable measures, implemented consistently across all study participants?** *[PI: Important measures may be listed separately. PI may need to establish a threshold for what would constitute acceptable measures based on study topic. When subjective or objective measures could be collected, subjective measures based on selfreport may be considered as being less reliable and valid than objective measures such as clinical reports and lab findings. Replicate question when needed.]*	**Yes.....** **No.....** **Cannot determine or measurement approach not reported** **.....** **Explanation for rating:**
15. **Source of Information for Outcomes**	**Are outcomes assessed using valid and reliable measures, implemented consistently across all study participants?**	**Yes.....** **No.....** **Cannot determine or measurement approach**

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	[PI: Primary outcomes should be identified for abstractors and if there is more than one, they may be listed separately. Also, identify any relevant secondary outcomes and harms. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such as clinical reports and lab findings. Note for case-control studies: consider whether the ascertainment of cases was independent of exposure.]	**not reported** **.....** **Explanation for rating:**
16. **Equality of Length of Follow-Up for Participants**	**Is the length of follow-up the same for all groups?** *[For case-control studies, are cases and controls matched on length of followup? Abstractor: When follow-up was the same for all study participants, the answer is yes. If different lengths of follow-up were adjusted by statistical techniques, (e.g., survival analysis), the answer is yes. Studies in which differences in follow-up were ignored should be answered no.]*	**Yes.....** **No or cannot determine** **.....** **Not0 applicable: cross-sectional or only one group followed over time.....** **Explanation for rating:**
17. **Length of Follow-up Adequate**	**Is the length of time following the intervention/exposure sufficient to support the evaluation of primary outcomes and harms?** *[PI: Primary outcomes (including harms) should be identified for abstractors. Important measures may be listed separately. Abstractors should be provided with specific criteria for sufficient length of follow-up based on prior research or theory. Drop if entire body of evidence is cross-sectional or if minimal length of follow-up period is specified through inclusion criteria.]*	**Yes.....** **Partially: some primary outcomes are followed for a sufficient length of time.....** **No.....** **Cannot determine** **.....** **Not applicable: cross-sectional** **.....** **Explanation for rating:**
18. **Completeness of Follow-Up**	**Did attrition from any group exceed [x] percent?** *[PI: Attrition is measured in relation to the time between baseline (allocation in some instances) and outcome measurement for both retrospective and prospective studies and could include data loss from crossover. Attrition rates may vary by outcome and time of measurement. Specify the criterion to meet relevant standards for the topic. Specify measurement period of interest, if repeated measures. Cochrane standard for attrition is 20 percent for shorter term (<1 year) and 30 percent for longer term (≥ 1 year). Drop if entire body of evidence is cross-sectional]*	**Yes.....** **No.....** **Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional.....** **Explanation for rating:**
19. **Attrition difference**	**Did attrition differ between groups by more than 20 percent?**	**Yes.....** **No.....**

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	[PI: If appropriate, modify difference criterion to meet relevant standards for the topic. Attrition rates may vary by outcome and time of measurement. Drop if entire body of evidence is cross-sectional or case series.]	**Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional or only one group followed—case series, one-arm study.....** **Explanation for rating:**
20. **Assessment of Baseline Comparability**	**Does the analysis control for baseline differences between groups?** *[PI: Drop if entire body of evidence is case series or case control. Define adequate control. List critical baseline differences that need to be controlled.]*	**Yes.....** **No.....** **Insufficient reporting to be able to determine .....** **Not applicable: only one group, no comparison group (case series), or case control study, no difference in measured baseline characteristics .....** **Explanation for rating:**
21. **Identification of Prognostic Factors (Effect Modifiers and Confounders)**	**Are confounding and/or effect modifying variables assessed using valid and reliable measures across all study participants?** *[PI: Some characteristics may require that sources for establishing their validity and/or reliability be described or referenced. If so, provide instruction to abstractors.]*	**Yes.....** **No.....** **Cannot determine or source for measures not reported.....** **Not applicable: no confounders or effect modifiers included in the study.....** **Explanation for rating:**
22. **Case-Mix Adjustment**	**Were the important confounding and effect modifying variables taken into account in the design and/or analysis (e.g., through matching, stratification, interaction terms, multivariate analysis, or other statistical adjustment)?** *[PI: Provide instruction to abstractors on adequate adjustment for confounding and testing for effect modification.]*	**Yes.....** **Partially: some variables taken into account or adjustment achieved to some extent.....** **No: not accounted for or not identified.....** **Cannot determine .....** **Explanation for rating:**
23. **Intention-to-Treat Analysis**	**In cases of high loss to follow-up (or differential loss to follow-up), is the impact assessed (e.g., through sensitivity**	**Yes.....** **No.....** **Cannot determine**

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	analysis or other adjustment method)?	**.....** **Not applicable: no loss to follow-up or loss to follow-up was not considered to be high, crosssectional study, or case-control study selected on outcome** **.....** **Explanation for rating:**
24. **Primary outcomes missing**	**Are any important primary outcomes missing from the results?** *[PI: Identify all primary outcomes, including timing of measurement, that one would expect to be reported in the study.]*	**Yes.....** **No.....** **Cannot determine** **.....** **Explanation for rating:**
27. **Appropriate statistics for outcome**	***Are the statistical methods used to assess the main harm or adverse event outcomes appropriate to the data?*** *[Abstractor: Question relates to precision and may not be relevant for systematic reviews that are able to pool data. The statistical techniques used must be appropriate to the data and take into account issues such as controlling for dose-response, small sample size, clustering, rare outcomes, and multiple comparisons. In normally distributed data, the standard error, standard deviation, or confidence intervals should be reported. In non-normally distributed data, inter-quartile range should be reported.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Explanation for rating:**
28. **Appropriate interpretation**	**Are results believable taking study limitations into consideration?** *[Abstractor: This question is intended to capture the overall quality of the study. Consider issues that may limit your ability to interpret the results of the study. Review responses to earlier questions for specific criteria.]*	**Yes.....** **Partially.....** **.** **No.....** **Explanation for rating:**
29. **Funding**	**Is the source of funding identified?** *[PI: The relevance of this question will depend upon the topic. This question may be modified to identify particular sources of funding (e.g., industry, government, university, or foundation funding).]*	**Yes.....** **No.....** **Explanation for rating:**

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Contributors: ZX conceived the study and drafted the protocol. LY and GL revised it. ZX, GY, YQ, ZC and ZW developed the search strategies and will run it. ZX and will select studies and extract data. ZX and ZC will analyze the data. All authors have approved the final edition of the protocol.

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Competing interests: None declared.

Ethical approval: This study does not require ethics approval.

Data sharing: No additional data are available

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Correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation : a protocol of systematic review and meta-analysis

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Correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation: a protocol of systematic review and meta-analysis

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Keywords: blood pressure, masked hypertension, endothelial dysfunction, flow-mediated dilation, meta-analysis protocol

Word count: 3071

Abstract

Introduction

A surrogate marker to evaluate artery endothelial response when stimulated by reactive hyperemia, known as brachial flow-mediated dilation (FMD), has prognostic value for predicting hypertensive organ damage and cardiovascular disease (CVD) events. However, the degree of correlation between brachial FMD and masked hypertension (MH) outcomes is still unclear. Therefore, this study's purpose is to pool data regarding FMD with respect to masked hypertension.

Methods and analysis

Electronic databases, MEDLINE, EMBASE, China National Knowledge Infrastructure (CNKI), and the Cochrane Library, will be searched for keywords endothelial dysfunction, flow-mediated dilation and masked hypertension, masked uncontrolled hypertension (MUCH), and prehypertension. Eligibility criteria are as shown: population—adults (18 years old or older) without hypertension at baseline, with suspected endothelial dysfunction, or from MH/MUCH populations (office BP <140/90 mmHg and home BP ≥135 mmHg and/or 85 mmHg) and from controlled clinical trials, cohort studies, or randomized and controlled trials; exposures—any metrics for FMD; comparisons—participants without MH or MUCH; and outcome—change in FMD between the case group and control group. Two writers will be engaged in screening and collecting data independently; disagreements will be resolved through discussion. Data extraction will include primary data designated as hazard ratios, odds ratios, correlations, regression coefficients. Comprehensive Meta-Analysis Version 2.0 will be used to conduct related subgroup and sensitivity analyses and publication bias.

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Ethics and dissemination

This study does not require ethics approval. It will be submitted to a peer-reviewed journal.

PROSPERO registration number: CRD42020208362

Strengths and limitations of this study

- This is the first meta-analysis we are aware of to report the correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation.
- The study will evaluate a new method diagnosing masked hypertension or masked uncontrolled hypertension.
- Since we include different types of studies and studies in various regions worldwide, relevant bias cannot be avoided.

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Background

Masked hypertension is defined as possessing a normal blood pressure in the office and an elevated measurement outside of it[1]. Diagnosing MH still has inherent challenges. Both missed and excess diagnoses have undesirable consequences. Undiagnosed MH could frequently develop into damage at targeted organs prior to transitioning to sustained hypertension[2]. Masked control of hypertension is more common in patients with hypertension, with a prevalence rate of 30-50%[3]. MH described as normal office blood pressure (BP) and a 24-hour mean increase in ambulatory blood pressure monitoring (ABPM) have been associated with CVD. Moreover, it was discovered that taking antihypertensive medication, defined by masked uncontrolled hypertension (MUCH), has an increased prevalence rate for masked hypertension compared to those not receiving antihypertensive treatment[4]. A definitive and accurate diagnosis is therefore required for all patients with suspected MH. Although studies have shown that hypertension is associated with an increased risk of CVD, the actual concern surrounding this health risk is not the mild elevation in BP that may be measured in a clinic, but masked hypertension being underestimated.

The diagnostic strategy starts with a clinical probability assessment and measurement of BP in the clinic. In patients with a high/likely clinical probability, ABPM and home blood pressure monitoring (HBPM) are required. ABPM is currently the predominant test for MH. The test has been universally verified in terms of diagnostic accuracy and outcome studies[5-7]. However, ABPM has some limitations. It is inconvenient compared with HBPM[8], difficult to operate, and with constant concerns about possible overdiagnosing and overtreating hypertension.

In recent years, the technology around quantifying endothelial function has rapidly evolved, which allowed for the introduction of flow-mediated dilation, a non-invasive method of assessing endothelial dysfunction[9]. FMD has been reported to be a predictor for hypertensive organ damage and CVD events[10 11], largely replacing the invasive method administered during cardiac catheterization. Many studies aimed at assessing the correlation between FMD and CVD risk. More recently, a few systematic reviews and meta-analyses have been published and concluded that FMD is positively associated with a risk for future cardiovascular events[12 13]. However, the clinical community remains uncertain about the relationship between MH or MUCH diagnosed through ABPM and flow-mediated vasodilation responses. Indeed, the concrete relationship that exists between FMD and MH or MUCH remains uncertain. For example, there is a bias inherent in FMD results in some studies, thus influencing the ultimate diagnosis conclusion. Therefore, although previously published meta-analyses focusing on brachial FMD have been reported for CVD outcomes, summarized results related to MH or MUCH are still lacking.

Methods

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Aims

The study aims to synthesize present reports regarding FMD and masked hypertension. The reporting of this study protocol will adhere to the PRISMA-P guidelines (supplementary appendix 1) and was registered on the PROSPERO database[14]. Any changes to this study will be updated on PROSPERO.

Search strategy

We will search for relevant articles in English and Chinese in these databases: MEDLINE, EMBASE, CNKI, and the Cochrane Library. The search period will be from the date when each database was established to November 2020. In addition, we will perform a manual search of related references except from the electronic search. We will contact the authors to obtain more information if necessary. Gray literature and unpublished research will be excluded. The MEDLINE search strategy is presented in Table 1.

Table 1 Search String

MEDLINE
"flow-mediated vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) dependent vasodilation (or vasodilatation or dilation or dilatation) or endothelial (or endothelium) function (or dysfunction), or FMD or vascular reactivity, and brachial artery" AND "cardiovascular diseases (or coronary disease) or masked hypertension (or hypertension or white coat hypertension or prehypertension or masked uncontrolled hypertension)"

Eligibility criteria

Population: adults (18 years old or older) without hypertension at baseline, with suspected endothelial dysfunction, from MH/MUCH populations (office BP <140/90 mmHg and home BP ≥135 mmHg and/or 85 mmHg) [15]

Exposure: any measurement of FMD utilizing any general metric.

Comparator/control: participants without MH or MUCH, including participants with normotension, white-coat hypertension, prehypertension, sustained hypertension, etc.

Outcomes: change in FMD between the case group and control group, including FMD magnitude expressed as percentage change from baseline, integrated FMD response calculated as the area under the dilation curve, etc.

Study design

Experimental studies designed as randomized controlled trials and non-randomized trials, observational studies designed as case-control studies and longitudinal cohort will be eligible. Prospective and retrospective studies will also be included. We will exclude any cross-sectional studies, case reports, and series.

Exclusion

Studies adopting patient self-reporting to determine masked hypertension, including self-reporting in a physician's diagnosis, which is not further verified consistently with the diagnostic criteria defined by the study outcomes are excluded.

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Study selection process

All citations identified through our search strategy will be imported into an EndNote library; any duplicates will be eliminated. First, two review authors will select the titles and abstracts from studies by searching and identifying potentially eligible studies independently. Then they will check against the inclusion criteria. Any disagreements at all stages during the selection process will be resolved through consultation or consensus with the corresponding author. Finally, a PRISMA flow chart will provide selection process and the reasons for any exclusions (as shown in Figure1).

Data extraction

After selecting initial studies, eligibility information will be extracted for each study relating to study identification (authors, years of publication, countries where the studies took place, and other information), study design and study characteristics (e.g. sample size, adverse events, duration of follow-ups), characteristics of the populations being studies (age, sex, BMI, dyslipidemia, smoking and stroke history, diabetes, kidney disease, liver disease and coronary heart disease), and methods of measuring BP. Any collected primary outcome data will include office BP (systolic and diastolic blood pressure), 24-hour ABPM (systolic and diastolic blood pressure), hypertension phenotypes, and brachial artery FMD outcomes, reported as categorical numbers. One writer for all studies will extract this data, which will be checked by the other writer to reduce human error. All disagreements will be resolved by the two writers.

Risk of bias

We will utilize the RTI item bank to evaluate methodological bias upon all included studies (supplementary appendix 2)[16].

Patient and public involvement

No patients and members of the public will be directly involved. Only data that already exists in the literature and the sources mentioned above will be used for this study.

Data Synthesis

We will present a detailed description of the results for all included studies in both text and table lists. We will describe these studies that are qualitatively relevant to study identifications (authors, years of publication, countries where the studies took place, and other information), study designs, and study characteristics (experimental or observational, sample size, adverse events, duration of follow-ups), patients (age, sex, BMI), the methods used to measure different blood pressures, and the values of FMD.

Meta-analysis

Comprehensive Meta-Analysis Version 2.0 will be used to conduct the meta-analyses. The summary effect measures may include hazard ratios, odds ratios, correlations regression coefficients, etc. When data is pooled together, we will use a fixed-effects model. In case a significant heterogeneity appears, we will use a random-effects model to make an estimate of effect size. If the characteristics of included studies are

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extremely heterogeneous, we will not pool results. Instead, we will make a narrative table listing the related findings by context description.

We will utilize the χ^2 test to evaluate statistical heterogeneity and the I^2 statistic to estimate the amount of heterogeneity. We will consider a value of $p < 0.1$ as heterogeneity in χ^2 test, whereas, according to the Cochrane Handbook for Systematic Reviews[17 18], when I^2 is among 0–60% heterogeneity is regarded as moderately unimportant; otherwise, it indicates substantial heterogeneity [19]. We will test reasons for heterogeneity through subgroup analysis.

Subgroup and sensitivity analyses

Considering the possibility of using different covariates in the study, we will conduct a subgroup analysis to assess the impact of each study feature on the study's primary outcome. We will conduct a subgroup analysis of patients with MH and MUCH. Also, subgroup analyses will be conducted for combined masked hypertension outcomes stratified by sex and age if possible. Study setting (rural vs urban or developed vs developing country) will also be considered. Sensitivity analyses will be implemented with the effects of age < 50 and > 50 years old, or as mean or median age in meta-regression alternatively, due to varied incidence rates of hypertension in different ages[20].

Assessment of publication bias

Egger's regression asymmetry method will be used to evaluate the publication bias[21].

GRADE framework for quality of evidence

The included study will use the Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidelines[22] to evaluate the quality of evidence and strength of recommendations.

Discussion

The purpose of this study is intended to provide the latest evidence for the risk of masked hypertension and endothelial dysfunction measured by FMD. Our study will contribute to this field by clarifying whether FMD, derived from ultrasound imaging, is associated with these blood pressure outcomes. It is of the actual, consistent evidence that there is an increased risk of cardiovascular disease related to masked hypertension[23]. The findings might also inform clinical physicians and policymakers with respect to hypertension management, especially when diagnosing masked hypertension.

There are a few limitations that will weaken the findings and applications of the study because endothelial dysfunction is the result of multiple interactions between cardiovascular regulatory function and environmental and behavioral factors[24]. As such, we are uncapable of discovering antecedent risk factors for higher FMD or ways to modulate and lessen masked hypertension risks. Similarly, FMD is interdependent with blood pressure in some forms; thus, it may be challenging to illustrate the effects of FMD independent from blood pressure. The included studies will also probably measure blood pressure by different methods, which could cause methodological heterogeneity. Limitations will also be associated with the outcomes of masked

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hypertension based on varying levels of effectiveness and heterogeneity. The ability to correctly judge the prognosis of masked hypertension outcomes will be related to the participants' ages, follow-up durations, and diseases. Furthermore, original studies may also include study heterogeneity and a high risk of bias, thus limiting the final conclusion drawn in this field. Moreover, as the included studies will only involve the English and Chinese languages, the generalizability of the study results published in other languages and other settings is finite in measurement.

In conclusion, given there are no studies available that assess endothelial functionality measured by FMD within the scope of masked hypertension, there is a pressing need to perform a meta-analysis on how FMD performs in diagnosing masked hypertension. The study will help summarize the available research evidence, and the findings may have guiding significance for clinical practices and diagnosing hypertension.

Contributors: ZW proposed the conception and acquired funding. LY and ZC are involved in the study design and project administration. ZX, GL, and ZC drafted the manuscript.

Competing interests: None declared.

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Data sharing statement: Not applicable.

References

1. Anstey DE, Pugliese D, Abdalla M, et al. An Update on Masked Hypertension. *Current hypertension reports* 2017;19(12):94. doi: 10.1007/s11906-017-0792-4 [published Online First: 2017/10/27]

2. Franklin SS, O'Brien E, Staessen JA. Masked Hypertension: Understanding Its Complexity. *European heart journal* 2017;38(15):1112-18. doi: 10.1093/eurheartj/ehw502 [published Online First: 2016/11/12]

3. Banegas JR, Ruilope LM, de la Sierra A, et al. High Prevalence of Masked Uncontrolled Hypertension in People with Treated Hypertension. *Eur Heart J* 2014;35(46):3304-12. doi: 10.1093/eurheartj/ehu016 [published Online First: 2014/02/06]

For peer review only

4. Franklin SS, Thijs L, Li Y, et al. Masked Hypertension in Diabetes Mellitus: Treatment Implications for Clinical Practice. *Hypertension (Dallas, Tex : 1979)* 2013;61(5):964-71. doi: 10.1161/hypertensionaha.111.00289 [published Online First: 2013/03/13]

5. Gaborieau V, Delarche N, Gosse P. Ambulatory Blood Pressure Monitoring Versus Self-measurement of Blood Pressure at Home: Correlation with Target Organ Damage. *Journal of Hypertension* 2008;26(10):1919-27. doi: 10.1097/HJH.0b013e32830c4368 [published Online First: 2008/09/23]

6. Conen D, Bamberg F. Noninvasive 24-h Ambulatory Blood Pressure and Cardiovascular Disease: a Systematic Review and Meta-analysis. *Journal of Hypertension* 2008;26(7):1290-9. doi: 10.1097/HJH.0b013e3282f97854 [published Online First: 2008/06/14]

7. Zhang L, Li Y, Wei FF, et al. Strategies for Classifying Patients Based on Office, Home, and Ambulatory Blood Pressure Measurement. *Hypertension (Dallas, Tex : 1979)* 2015;65(6):1258-65. doi: 10.1161/hypertensionaha.114.05038 [published Online First: 2015/04/15]

8. Stergiou GS, Salgami EV, Tzamouranis DG, et al. Masked Hypertension Assessed by Ambulatory Blood Pressure Versus Home Blood Pressure Monitoring: is It the Same Phenomenon? *American Journal of Hypertension* 2005;18(6):772-8. doi: 10.1016/j.amjhyper.2005.01.003 [published Online First: 2005/06/01]

9. Celermajer DS, Sorensen KE, Gooch VM, et al. Non-invasive Detection of Endothelial Dysfunction in Children and Adults at Risk of Atherosclerosis. *Lancet (London, England)* 1992;340(8828):1111-5. doi: 10.1016/0140-6736(92)93147-f [published

For peer review only

Online First: 1992/11/07]

10. Korkmaz H, Onalan O. Evaluation of Endothelial Dysfunction: Flow-mediated Dilation. *Endothelium : Journal of Endothelial Cell Research* 2008;15(4):157-63. doi: 10.1080/10623320802228872 [published Online First: 2008/07/30]

11. Inoue T, Matsuoka H, Higashi Y, et al. Flow-mediated Vasodilation As a Diagnostic Modality for Vascular Failure. *Hypertension Research : Official Journal of the Japanese Society of Hypertension* 2008;31(12):2105-13. doi: 10.1291/hypres.31.2105 [published Online First: 2009/01/14]

12. Frick M, Suessenbacher A, Alber HF, et al. Prognostic Value of Brachial Artery Endothelial Function and Wall Thickness. *Journal of the American College of Cardiology* 2005;46(6):1006-10. doi: 10.1016/j.jacc.2005.05.070 [published Online First: 2005/09/20]

13. Fathi R, Haluska B, Isbel N, et al. The Relative Importance of Vascular Structure and Function in Predicting Cardiovascular Events. *Journal of the American College of Cardiology* 2004;43(4):616-23. doi: 10.1016/j.jacc.2003.09.042 [published Online First: 2004/02/21]

14. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA Statement for Reporting Systematic Reviews and Meta-analyses of Studies That Evaluate Health Care Interventions: Explanation and Elaboration. *PLoS Medicine* 2009;6(7):e1000100. doi: 10.1371/journal.pmed.1000100 [published Online First: 2009/07/22]

15. Kabutoya T, Hoshide S, Ogata Y, Eguchi K, Kario K. Masked hypertension defined by home blood pressure monitoring is associated with impaired flow-mediated

For peer review only

vasodilatation in patients with cardiovascular risk factors. J Clin Hypertens (Greenwich). 2013 Sep;15(9):630-6. doi: 10.1111/jch.12155. Epub 2013 Jun 25. PMID: 24034655.

16. Viswanathan M, Berkman ND, Dryden DM, et al. AHRQ Methods for Effective Health Care. Assessing Risk of Bias and Confounding in Observational Studies of Interventions or Exposures: Further Development of the RTI Item Bank. Rockville (MD): Agency for Healthcare Research and Quality (US) 2013.

17. Moreira MT, Smith LA, Foxcroft D. Social Norms Interventions to Reduce Alcohol Misuse in University or College Students. *The Cochrane Database of Systematic Reviews* 2009(3):Cd006748. doi: 10.1002/14651858.CD006748.pub2 [published Online First: 2009/07/10]

18. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in medicine* 2002;21(11):1539-58. doi: 10.1002/sim.1186 [published Online First: 2002/07/12]

19. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring Inconsistency in Meta-analyses. *BMJ (Clinical research ed)* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

20. Takase H, Tanaka T, Takayama S, et al. Recent Changes in Blood Pressure Levels, Hypertension Prevalence and Treatment Rates, and the Rate of Reaching Target Blood Pressure in the Elderly. *Medicine* 2017;96(50):e9116. doi: 10.1097/md.0000000000009116 [published Online First: 2018/02/03]

21. Egger M, Davey Smith G, Schneider M, et al. Bias in Meta-analysis Detected by a Simple,

For peer review only

Graphical Test. *BMJ (Clinical research ed)* 1997;315(7109):629-34. doi: 10.1136/bmj.315.7109.629 [published Online First: 1997/10/06]

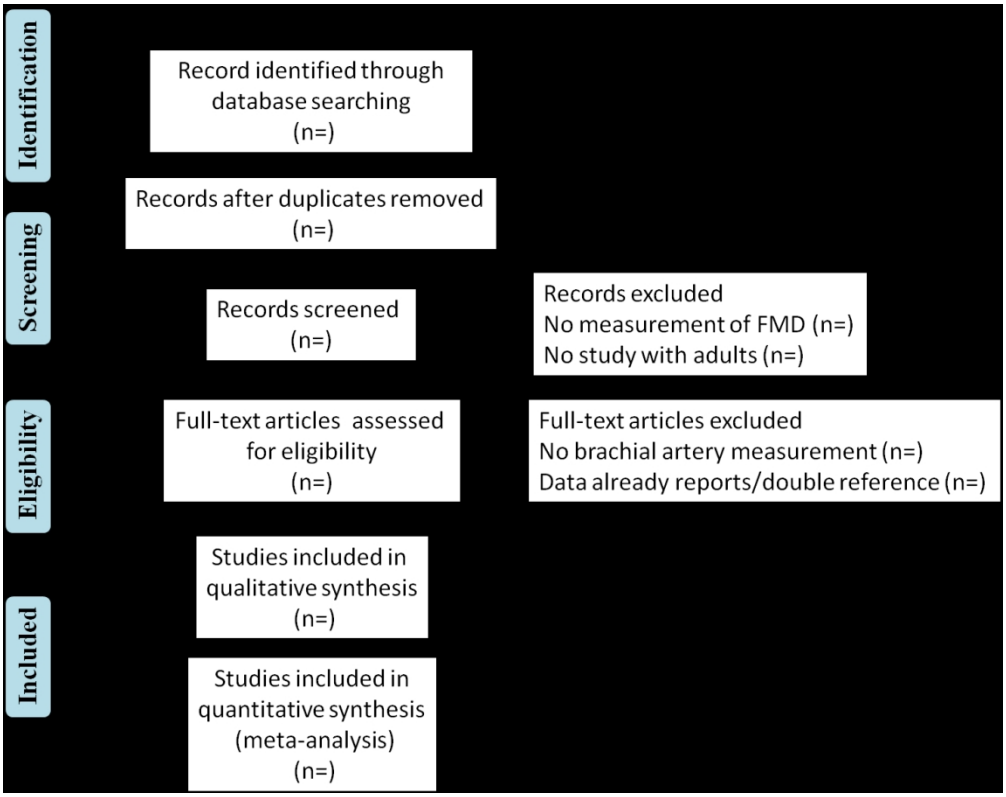
22. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: An Emerging Consensus on Rating Quality of Evidence and Strength of Recommendations. *BMJ (Clinical Research Ed)* 2008;336(7650):924-6. doi: 10.1136/bmj.39489.470347.AD [published Online First: 2008/04/26]

23. Pierdomenico SD, Cuccurullo F. Prognostic Value of White-coat and Masked Hypertension Diagnosed by Ambulatory Monitoring in Initially Untreated Subjects: An Updated Meta Analysis. *American Journal of Hypertension* 2011;24(1):52-8. doi: 10.1038/ajh.2010.203 [published Online First: 2010/09/18]

24. Charakida M, Masi S, Lüscher TF, et al. Assessment of Atherosclerosis: The Role of Flow-mediated Dilatation. *European Heart Journal* 2010;31(23):2854-61. doi: 10.1093/eurheartj/ehq340 [published Online First: 2010/09/25]

Figure 1 Flow chart of the study selection procedure

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248x195mm (150 x 150 DPI)

Appendix 1

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review		√	
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		√	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	√		32
Authors:					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	√		
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	√		
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		√	
Support:					
Sources	5a	Indicate sources of financial or other support for the review		√	
Sponsor	5b	Provide name for the review funder and/or sponsor		√	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		√	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	√		46-89
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	√		107-117
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	√		119-122
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	√		98-104
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	√		Table 1
Study records:					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	√		164-176
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	√		128-131
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	√		131-134
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	√		136-142
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	√		142-147
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the	√		150-151

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		outcome or study level, or both; state how this information will be used in data synthesis			
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	√		158-162
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	√		173-176
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	√		178-184
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned		√	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	√		186
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	√		188-190

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Appendix 2

RTI Risk of Bias Item Bank

Risk of bias item	Description	
1. **Retrospective/Prospective**	**Is the study design prospective, retrospective, or mixed?** *[Abstractor: Prospective design requires that the outcome has not occurred at the time the study is initiated and information is collected over time to assess relationships with the outcome (and includes nested case-control studies). Mixed design includes case-control or cohort studies in which one group is studied prospectively and the other retrospectively. A retrospective design analyzes data from past records. The question is not applicable to cross-sectional studies.]*	**Prospective..... Mixed..... Retrospective..... Cannot determine/not applicable..... Explanation for rating:**
2. **Inclusion/Exclusion Criteria**	**Are critical inclusion/exclusion criteria clearly stated (does not require the reader to infer)?** *[Principal Investigator (PI): Provide direction to abstractors by listing individual criteria of a priori significance and minimal requirements for criteria to be considered "clearly stated." Include this question to identify specific inclusion/exclusion criteria that should be consistently recorded across studies] [Abstractor: Use "Partially" if only some criteria are stated or if some criteria are not clearly stated (corresponding to directions provided by the PI). Note that studies may describe inclusion criteria alone (i.e., include x), exclusion criteria (i.e., do not include x), or a combination of inclusion and exclusion criteria.]*	**Yes..... Partially: some, but not all, criteria stated or some criteria not clearly stated..... No..... Explanation for rating:**
3. **Inclusion/Exclusion Criteria**	**Are the inclusion/exclusion criteria measured using valid and reliable measures?** *[PI: Separately specify each criterion that abstractors should consider based on its relevance to study bias. It is unlikely that all criteria will need to be evaluated in relation to this question. Provide direction to abstractors on valid and reliable measurement of each criterion that is to be considered. For example, prior exposure or disease status is a frequent inclusion/exclusion criterion, particularly in inception cohorts. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such*	**Yes..... No..... Cannot determine; measurement approach not reported..... Explanation for rating:**

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	as clinical reports and lab findings. Replicate question to evaluate each individual inclusion/exclusion criterion.]	
4. **Inclusion/Exclusion Criteria**	**Did the study apply inclusion/exclusion criteria uniformly to all comparison groups/arms of the study?** *[PI: Drop question if not relevant to entire body of evidence (e.g., all case-series, singlearm studies).]*	**Yes..... Partially: some, but not all criteria, applied to all arms or not clearly stated if some criteria are applied to all arms ..... No..... Cannot determine: article does not specify..... Not applicable: study has only one arm and so does not include comparison groups..... Explanation for rating:**
5. **Recruitment across groups**	**Was the strategy for recruiting participants into the study the same across study groups/arms of the study?** ***[PIs: This question is likely to be more relevant for prospective or mixed designs than retrospective designs. Drop question if not relevant to entire body of evidence (e.g., all studies generally have only one arm).]***	**Yes..... No..... Cannot determine ..... Not applicable: one study group/arm ..... Explanation for rating:**
6. **Statistical power**	**Was the sample size sufficiently large to detect a clinically significant difference of 5% or more between groups in at least one primary outcome measure?** *[PI: Specify a different percent, if clinically relevant for each outcome of interest. Question relates to precision; reviewers whose evaluation of quality is limited to considerations of systematic error or risk of bias (not random error/precision) need not include this question. Reviewers who include both precision and systematic error in their evaluation of quality but rely on meta-analysis for pooled estimates need not include this question. PIs who choose to include considerations of precision in their assessment may include the question, but should be aware of the need for collaboration between clinical and statistical expertise in determining the threshold for a clinically adequate sample size.]*	**Yes..... No..... Explanation for rating:**
7.	***What is the level of detail in describing***	**High: very clear, all PI-**

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Detail of exposure	***the intervention or exposure?*** *[PI: Specify which details need to be stated (e.g., intensity, duration, frequency, route, setting, and timing of intervention/exposure). For case-control studies, consider whether the condition, timing, frequency, and setting of symptoms are provided in the case definition. PI needs to establish criteria for high, medium, or low response.]*	**required details provided** **Medium: somewhat clear, majority of PI required details provided** **.....** **Low: unclear, many PI-required details missing** **.....** **Explanation for rating:**
8. **Specification of outcomes**	**Are the important outcomes prespecified by the researchers? Do not consider harms in answering this question unless they should have been pre-specified.** *[PI: This question can be asked for all outcomes together or replicated for each event. Each adverse event of interest should be specified for abstractors. Relevant source information includes all study data, including what may have been established in relation to an initial randomized controlled trial. Drop question if not relevant (e.g., primary outcome for case-control studies).]*	**Yes** **.....** **Partially** **.....** **No** **.....** **Not applicable** **.....** **Explanation for rating:**
9. **Appropriate comparison group**	**Is the selection of the comparison group appropriate, after taking into account feasibility and ethical considerations.** *[PI: Provide instruction to the abstractor based on the type of study. Interventions with community components are likely to have contamination if all groups are drawn from the same community. Interventions without community components should select groups from the same source (e.g., community or hospital) to reduce baseline differences across groups. For case-control studies, controls should represent the population from which cases arose; that is, controls should have met the case definition if they had the outcome.]*	**Yes.....** **No.....** **Cannot determine or no description of the derivation of the comparison group** **.....** **Not applicable: study does not include a comparison group (case series, one study arm) .....** **Explanation for rating:**
10. **Attempt To Balance**	**Any attempt to balance the allocation between the groups (e.g., through stratification, matching, propensity scores).** *[PI: This is most likely to be used in case-control study designs. Drop if not relevant to the body of evidence.]*	**Yes or study accounts for imbalance between groups through a post hoc approach such as multivariate analysis** **.....** **No or cannot determine** **.....** **Not applicable: study does not include a comparison group (case**

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		series or one study arm) **Explanation for rating:**
11. **Contamination**	**Did researchers isolate the impact from a concurrent intervention or an unintended exposure that might bias results, e.g., through multivariate analysis, stratification, or subgroup analysis?** *[PI: specify interventions or exposures for abstractors.]*	**Yes.....** **Partially.....** **.** **No or do not know: concurrent intervention or Unintended exposure is not described)** **.....** **Not applicable: no concurrent interventions or unintended exposures likely.....** **Explanation for rating:**
12. **Variation in execution of protocol**	**Did execution of the study vary from the intervention protocol proposed by the investigators and therefore compromise the conclusions of the study?** *[PI: Consider intensity, duration, frequency, route, setting, and timing of intervention/exposures. Drop if not relevant for body of literature.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Not applicable: not an intervention study** **.....** **Explanation for rating:**
13. **Blind Outcomes Assessment**	**Were the outcome assessors blinded to the intervention or exposure status of participants?** *[PI: There may be circumstances where clinical evaluators cannot be blinded to exposure status. Drop if not relevant to the body of literature.]*	**Yes.....** **No.....** **Not applicable: assessor cannot be blinded.....** **Explanation for rating:**
14. **Exposures assessed using valid**	**Are interventions/exposures assessed using valid and reliable measures, implemented consistently across all study participants?** *[PI: Important measures may be listed separately. PI may need to establish a threshold for what would constitute acceptable measures based on study topic. When subjective or objective measures could be collected, subjective measures based on selfreport may be considered as being less reliable and valid than objective measures such as clinical reports and lab findings. Replicate question when needed.]*	**Yes.....** **No.....** **Cannot determine or measurement approach not reported** **.....** **Explanation for rating:**
15. **Source of Information for Outcomes**	**Are outcomes assessed using valid and reliable measures, implemented consistently across all study participants?**	**Yes.....** **No.....** **Cannot determine or measurement approach**

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	[PI: Primary outcomes should be identified for abstractors and if there is more than one, they may be listed separately. Also, identify any relevant secondary outcomes and harms. Subjective measures based on self-report tend to have lower reliability and validity than objective measures such as clinical reports and lab findings. Note for case-control studies: consider whether the ascertainment of cases was independent of exposure.]	**not reported** **.....** **Explanation for rating:**
16. **Equality of Length of Follow-Up for Participants**	**Is the length of follow-up the same for all groups?** *[For case-control studies, are cases and controls matched on length of followup? Abstractor: When follow-up was the same for all study participants, the answer is yes. If different lengths of follow-up were adjusted by statistical techniques, (e.g., survival analysis), the answer is yes. Studies in which differences in follow-up were ignored should be answered no.]*	**Yes.....** **No or cannot determine** **.....** **Not0 applicable: cross-sectional or only one group followed over time.....** **Explanation for rating:**
17. **Length of Follow-up Adequate**	**Is the length of time following the intervention/exposure sufficient to support the evaluation of primary outcomes and harms?** *[PI: Primary outcomes (including harms) should be identified for abstractors. Important measures may be listed separately. Abstractors should be provided with specific criteria for sufficient length of follow-up based on prior research or theory. Drop if entire body of evidence is cross-sectional or if minimal length of follow-up period is specified through inclusion criteria.]*	**Yes.....** **Partially: some primary outcomes are followed for a sufficient length of time.....** **No.....** **Cannot determine** **.....** **Not applicable: cross-sectional** **.....** **Explanation for rating:**
18. **Completeness of Follow-Up**	**Did attrition from any group exceed [x] percent?** *[PI: Attrition is measured in relation to the time between baseline (allocation in some instances) and outcome measurement for both retrospective and prospective studies and could include data loss from crossover. Attrition rates may vary by outcome and time of measurement. Specify the criterion to meet relevant standards for the topic. Specify measurement period of interest, if repeated measures. Cochrane standard for attrition is 20 percent for shorter term (<1 year) and 30 percent for longer term (≥ 1 year). Drop if entire body of evidence is cross-sectional]*	**Yes.....** **No.....** **Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional.....** **Explanation for rating:**
19. **Attrition difference**	**Did attrition differ between groups by more than 20 percent?**	**Yes.....** **No.....**

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	[PI: If appropriate, modify difference criterion to meet relevant standards for the topic. Attrition rates may vary by outcome and time of measurement. Drop if entire body of evidence is cross-sectional or case series.]	**Cannot determine: includes retrospective designs not stating number eligible at baseline .....** **Not applicable: cross-sectional or only one group followed—case series, one-arm study.....** **Explanation for rating:**
20. **Assessment of Baseline Comparability**	**Does the analysis control for baseline differences between groups?** *[PI: Drop if entire body of evidence is case series or case control. Define adequate control. List critical baseline differences that need to be controlled.]*	**Yes.....** **No.....** **Insufficient reporting to be able to determine .....** **Not applicable: only one group, no comparison group (case series), or case control study, no difference in measured baseline characteristics ..... .** **Explanation for rating:**
21. **Identification of Prognostic Factors (Effect Modifiers and Confounders)**	**Are confounding and/or effect modifying variables assessed using valid and reliable measures across all study participants?** *[PI: Some characteristics may require that sources for establishing their validity and/or reliability be described or referenced. If so, provide instruction to abstractors.]*	**Yes.....** **No.....** **Cannot determine or source for measures not reported..... ..** **Not applicable: no confounders or effect modifiers included in the study.....** **Explanation for rating:**
22. **Case-Mix Adjustment**	**Were the important confounding and effect modifying variables taken into account in the design and/or analysis (e.g., through matching, stratification, interaction terms, multivariate analysis, or other statistical adjustment)?** *[PI: Provide instruction to abstractors on adequate adjustment for confounding and testing for effect modification.]*	**Yes.....** **Partially: some variables taken into account or adjustment achieved to some extent.....** **No: not accounted for or not identified.....** **Cannot determine .....** **Explanation for rating:**
23. **Intention-to-Treat Analysis**	**In cases of high loss to follow-up (or differential loss to follow-up), is the impact assessed (e.g., through sensitivity**	**Yes.....** **No.....** **Cannot determine**

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	analysis or other adjustment method)?	**.....** **Not applicable: no loss to follow-up or loss to follow-up was not considered to be high, crosssectional study, or case-control study selected on outcome** **.....** **Explanation for rating:**
24. **Primary outcomes missing**	**Are any important primary outcomes missing from the results?** *[PI: Identify all primary outcomes, including timing of measurement, that one would expect to be reported in the study.]*	**Yes.....** **No.....** **Cannot determine** **.....** **Explanation for rating:**
27. **Appropriate statistics for outcome**	***Are the statistical methods used to assess the main harm or adverse event outcomes appropriate to the data?*** *[Abstractor: Question relates to precision and may not be relevant for systematic reviews that are able to pool data. The statistical techniques used must be appropriate to the data and take into account issues such as controlling for dose-response, small sample size, clustering, rare outcomes, and multiple comparisons. In normally distributed data, the standard error, standard deviation, or confidence intervals should be reported. In non-normally distributed data, inter-quartile range should be reported.]*	**Yes.....** **Partially.....** **.** **No.....** **Cannot determine** **.....** **Explanation for rating:**
28. **Appropriate interpretation**	**Are results believable taking study limitations into consideration?** *[Abstractor: This question is intended to capture the overall quality of the study. Consider issues that may limit your ability to interpret the results of the study. Review responses to earlier questions for specific criteria.]*	**Yes.....** **Partially.....** **.** **No.....** **Explanation for rating:**
29. **Funding**	**Is the source of funding identified?** *[PI: The relevance of this question will depend upon the topic. This question may be modified to identify particular sources of funding (e.g., industry, government, university, or foundation funding).]*	**Yes.....** **No.....** **Explanation for rating:**

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